

**Louis Racette**

**Visualizing Data in More Than Three  
Dimensions**

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**Advisors: Andreas Wilke, Daqing Hou**

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## Glossary/ Technical Terms

**Apache Tomcat:** Apache Tomcat is a server environment that allows client machines to access data on a single server machine.

**Java:** Java is a programming language used for writing the applications used in this report.

**Java3d:** Java3d is an API used for the n-dimensional grapher that allows three-dimensional modeling with java.

**N-Dimensional space:** This document refers to N-dimensional space in multiple sections. In this case, N is an arbitrary positive integer.

**Pearson Correlation:** This is a correlation statistic that provides the strength of the linear relationship between two variables, ranging from -1 (perfect negative correlation) to 0 (no correlation) to 1 (perfect positive correlation).

## Executive Summary

This paper contains development details for an n-dimensional graphing software program and the results of an experiment run to analyze the software's effectiveness in creating human-readable diagrams. This design and research project was conducted as an interdisciplinary honors thesis from the summer of 2011 to the fall of 2012. The graphing software was designed to create graphical representations of data in an arbitrary number of dimensions; testing of these representations was conducted during the spring and fall of 2012 using PY151 introductory psychology students.

## Introduction

Current data analysis techniques rely on the use of two or three dimensional comparisons to convey information. However, a single experiment can generate tens or hundreds of dimensions of data. For example, water quality analysis along a river would generate data involving three spatial dimensions, various oxide contents, dissolved gasses, biological activity, and so on. Representing this data in its full form, with all dimensions intact, would allow broader comparisons to be made and generalized patterns to be seen that would otherwise be obscured.

## Literature Review:

There have been many attempts to understand n-dimensional object representations and their significance to mathematics. The first recorded attempt is Edwin Abbot's *Flatland*, a mathematical treatise written as a fictional narrative (Abbot, 1884). This book explores the possibility of 0,1, and 2-dimensional universes, and contains the story of a sentient square taken into the third dimension by an omniscient sphere. In doing so, Abbot postulates that more than three dimensions could be thought of by analogy. From a conversation between the square and the cube:

*"In One Dimension, did not a moving Point produce a Line with TWO terminal points?"*

*In Two Dimensions, did not a moving Line produce a Square with FOUR terminal points?"*

*In Three Dimensions, did not a moving Square produce--did not this eye of mine behold it--that blessed Being, a Cube, with EIGHT terminal*

*points?*

*And in Four Dimensions shall not a moving Cube--alas, for Analogy, and alas for the Progress of Truth, if it be not so--shall not, I say, the motion of a divine Cube result in a still more divine Organization with SIXTEEN terminal points?"*

Abbot then continues this speculation to beyond four dimensions; this geometry-by-analogy makes the exploration of n-dimensional space possible.

Later, Albert Einstein made use of this concept of more than three dimensions to explore relativity and space-time. This is explained in Brian Greene's *The Elegant Universe* using multiple analogies; suffice it to say that four-dimensional mathematics is necessary for relativity to work (Greene, 2000).

Brian Coxeter's 1963 book, *Regular Polytopes*, explored n-dimensional geometry in detail, with specific reference to the platonic ideal shapes. Coxeter used angle and line segment measurements to determine the shape of hyper-objects without needing to visualize them (Coxeter, 1963).

Hand drawing of n-dimensional objects requires a great deal of effort and typically results in incomprehensible images. Computers, however, can quickly perform the calculations required to produce series of images or animations representing an n-dimensional object that can provide a deeper understanding than a hand drawing. Michael Knoll's paper, *A Computer Technique for Displaying N-Dimensional Hyper Objects*, explains how to use a computer to draw such objects using projections. Projections essentially simplify an n-dimensional object into real coordinate axes by repeatedly projecting into one lower dimension until the object is three-dimensional. From the paper: "*Any n-dimensional hyperobject could then be manipulated mathematically by a digital computer. The final three-dimensional projection... could be drawn automatically on a computer-controlled visual display device as a stereoscopic movie*" (Knoll, 1967, p. 1). Multiple projections can be done using either two or three dimensional axes sets. In these cases, comparisons between data dimensions are run exhaustively so that all possible comparisons are displayed. For example, a four-dimensional system with axes X,Y,Z and W would have 6 two-dimensional axes sets (XY, XZ, XW, YZ, YW, ZW) and 4 three-dimensional axes sets (XYZ, XYW, XZW, YZW).

In order to avoid this data loss through projection, the parallel-axis method was developed. Alfred Inselberg's 1985 paper, *The plane with parallel coordinates*, explores this representational method (Inselberg, 1985). Essentially, the parallel axis method represents points as lines connecting locations on axes; these axes are laid out as parallel to each other. This method preserves data in its entirety, but can be difficult to understand because it does not involve right-angle comparisons seen in traditional Euclidean geometry.

In 2004, Gary Felder laid out the basics of a different representational method in his instructional guide, *How to draw a five-dimensional cube*. This method, hereafter referred to as the Felder Method, is based

on “slices” in more than three dimensions. Each slice consists of the  $n$ -dimensional object’s projection into the lowest three axes at a specific value of the  $n$ th dimension. This creates a grid of three-dimensional axes with specified bounding values and increments (Felder, 2004).

Several psychological studies have been published on human navigation and search patterns in  $n$ -dimensional space. Unfortunately, most of these studies have been limited to four dimensional control of a three-dimensional viewing window. In 2001, Seyranian and D’Zmura published *Search and navigation in environments with four spatial dimensions*. This paper explored volunteers’ ability to navigate effectively through four-dimensional space (Seyranian, D’Zmura, 2001). In 2008, Alflalo and Graziano published *Four-dimensional spatial reasoning in humans*, exploring similar concepts of navigation and location in four dimensions (Alflalo, Graziano, 2008). In 2009, this idea was expanded using a 3-dimensional display (a cube consisting of inward-facing screens) in Wang et al.’s *Human four-dimensional spatial intuition in virtual reality* (Wang et al., 2009). All of these studies show similar results: given a significant training period, volunteers can begin to think in four dimensions. However, these navigation environments only displayed three dimensions at a time to participants- they relied on the projection method. Thus, from the studies provided, it is impossible to say if the participants were gaining intuitive understanding of four-dimensional mazes or simply learning to navigate in a novel environment. One possible explanation of participants’ acuity in learning four-dimensional mazes is the bi-coded spatial navigation theory. This theory posits that all human navigation takes place in two horizontal dimensions with some heuristics to represent the third vertical dimension. This means that the two horizontal dimensions are more accurately represented (in terms of distance and internal navigation) than the vertical dimension. Numerous animal studies and neurological evidence support this model (Jeffery et al., 2012). If this theory is accurate, then navigation in four or more dimensions could be accomplished using a similar set of learned heuristics.

This thesis project expands on this previous research by comparing the different visualization methods that have been previously formulated. Methods for drawing  $n$ -dimensional objects have been developed for mathematics and science without formal psychological testing of their effectiveness. Psychological testing has exclusively relied on a single three-dimensional projection method for data presentation. The thesis research comparatively tested the different visualization methods, allowing future psychological research to proceed with a better display method than the projection method currently used.

## Problem Statement

Thus far, psychological testing of  $n$ -dimensional understanding has been limited to projection methods. The alternative representation methods mentioned above may be more effective for conveying  $n$ -dimensional information, but they have not been empirically tested or compared. Thus, testing to determine the “best” representation method is necessary.



## Significance

Comparative testing of each of these representation methods will provide evidence for new data display algorithms. This will allow viewing of data in more than three dimensions; using these display models, patterns will be more visible than they would be using traditional two and three dimensional data analysis. Various algorithms already exist to simplify higher-dimensional data into more relatable forms; however, these algorithms (typically termed dimension reduction processes) can cause oversimplification and/or data loss. N-dimensional data presentation without this data loss could potentially aid in experimental analysis in all fields of science.

## Hypothesis

The best representation method for n-dimensional data would be expected to smoothly transition between axes sets. Thus, a method involving animation would be expected to perform better than a method involving a series of still images. For the experiment run, visualization method performance is quantified by the number of questions correctly answered for that method and the time required for participants to do so.

## Methodology

The thesis project involved two major components. First, software was written to draw three-dimensional representations of n-dimensional objects. This software was created using the java programming language and java3d plugin for graphics. The program stores graphical information as a set of points, where each point represents a set of n measurements for an n-dimensional system. Various display methods then utilize this point information to represent information graphically. This software can be used to generate still images or animations for each of these methods. Once generated, the various images were presented to volunteers in order to determine the most effective visualization method (within certain constraints). The methods initially chosen for testing were the parallel axis method, the Felder method, the multiple two-dimensional axes method, the multiple three-dimensional axes method, and the rotating axes method (where an optical illusion is used to switch between multiple three-dimensional spaces). Examples of each of each visualization method are shown in appendix A. After pilot testing with a group of six participants, the most effective methods were used in testing with a larger pool of participants. Pilot testing was conducted via a simple HTML webpage without any data gathering mechanisms; large scale testing was conducted with a specialized java applet that allowed data to be saved. This program collected participants' names, student numbers, and professor names along with their answers/reaction times for six mental rotation task questions.

### Question format:

Questions were designed to build on the research done by Roger Shepard on Mental Rotation. Shepard's original questions appeared as shown in figure 1; participants were asked to determine if two images shown side-by-side were the same object or different objects.

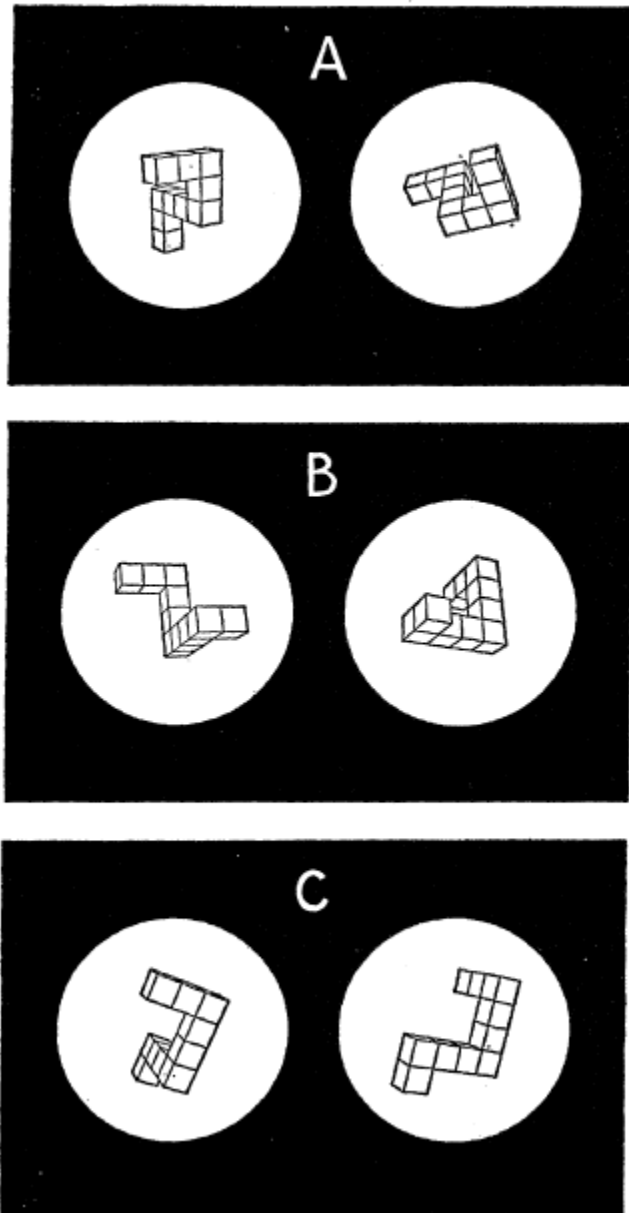


Figure 1: Rotation Images from Roger Shepard's experiments

Shepard (1971) hoped to quantify the processes underlying mental rotation and object manipulation using three-dimensional figures. He tested eight adult subjects with 1600 paired figures and measured

reaction time required to determine whether the two figures were identical, along with whether the participant was correct/incorrect. On average, only 3.2 % of responses were incorrect; within the correct responses, response time increased linearly relative to the angle of rotation between the two objects. In addition, this response time was unrelated to the axis of rotation- whether within the picture plane or perpendicular to it. These results suggest that mental rotation occurs at a maximum rate and that subjects are adept at mental rotation in three dimensions.

Replication of Shepard's experiment with children (Estes, 1998) and adolescents (Waber et al., 1982) revealed similar trends. Interestingly, male children tended to perform faster than female children on mental rotation tasks (Linn & Petersen, 1985).

Shepard's question structure was used for quantifying participant reactions to objects in more than three dimensions. This sort of task was well suited to the work required for participant testing, as it provided a measure of how well participants understood the information displayed to them in terms of both correct/incorrect answers and time required to respond.

## Results:

### Pilot Testing:

Pilot testing was conducted with six psychology undergraduates using five different questions types. All questions asked were in four-dimensional space. Five visualization methods were explored in this pilot survey: the parallel axes method (which relies on drawing all spatial dimensions parallel to each other, with points displayed as lines connecting each axis), the Felder method (which involves taking three-dimensional "slices" at regular intervals along each dimension higher than 3), the multiple 2d/3d axes methods (which display all possible two/three dimensional projection combinations for a given object) and the rotating axes method (which uses an optical illusion to transition between three-dimensional projections). These methods are shown in more detail in appendix A. The pilot survey format appeared as shown in appendix B. The first pilot survey had too few questions to provide any actionable data; the refined second pilot survey provided the results shown in table 1.

| Method        | Number correct per method<br>(out of 36 maximum) | Ranking |
|---------------|--------------------------------------------------|---------|
| Parallel      | 27                                               | 4       |
| Multiple 2D   | 30                                               | 3       |
| Multiple 3D   | 33                                               | 2       |
| Felder        | 16                                               | 5       |
| Rotating Axes | 36                                               | 1       |

Table 1: Response Summary for pilot testing (second round)

As can be seen for this rudimentary survey, the rotating axes method, multiple 3d axes method and multiple 2d axes method performed the best out of the five methods explored. The Felder method and parallel axes method performed worse. This is due largely to the data being visualized. The Felder method provides information only about discrete three-dimensional “slices” along a range of four-dimensional values; reconstructing the original four-dimensional object proved challenging for most participants- response rates for this visualization method was slightly below chance level. The parallel axes method provides information about individual points in four-dimensional space with a high degree of accuracy, but does a poor job of representing whole objects and interrelated points. The three successful methods relied upon two and three-dimensional visualization, which allowed participants to draw upon pre-existing navigation/visualization heuristics.

Based on this initial data, two methods were dropped from the later experiments: the parallel axes method and the Felder method. Both of these methods were relatively ineffective for conveying the information desired.

### Spring 2012 Testing:

Tests in the spring of 2012 were conducted using a refined experimental method that only focused on three data representation methods: the multiple three-dimensional, multiple two-dimensional and rotating axes methods. This experiment was conducted using the online survey system and images generated by the n-dimensional graphing software. See Appendix C for images from the spring survey. A total of 83 response sets were recorded by the software; 24 participants took the multiple two-dimensional axes method survey, 29 took the multiple three-dimensional survey and 30 took the rotating axes survey. The disparity in sample sizes is due to the nature of the online survey system- if a participant didn't complete his/her survey, then the system would record their data; this data was only rejected during analysis. Assignment to experimental groups was done sequentially- for every three participants, the first participant saw the multiple two-dimensional survey, the second the multiple three-dimensional survey, and the third the rotating axes survey. Question order for each participant was varied in a counterbalanced design, with the end result that no two participants saw the same survey. Results from all three surveys are summarized in table 2.

|                                                                                | Multiple 2D Method<br>(n = 24) |                       | Multiple 3D Method<br>(n = 29) |                       | Rotating Axes Method<br>(n=30) |                       |
|--------------------------------------------------------------------------------|--------------------------------|-----------------------|--------------------------------|-----------------------|--------------------------------|-----------------------|
|                                                                                | Mean                           | Standard<br>Deviation | Mean                           | Standard<br>Deviation | Mean                           | Standard<br>Deviation |
| Time<br>required for<br>survey<br>completion<br>(seconds)                      | 86.45                          | 84.43                 | 59.94                          | 21.01                 | 80.94                          | 23.38                 |
| Questions<br>Answered<br>Correctly<br>(Out of 6)                               | 4.625                          | 1.279                 | 4.862069                       | 1.3018                | 5.0667                         | 1.0483                |
| Pearson<br>Correlation<br>Between<br>Number<br>Correct and<br>Time<br>Required |                                | 0.28                  |                                | 0.0062                |                                | 0.15                  |

Table 2: Results from Spring 2012 testing

Unfortunately, most of the participants required less than three minutes to complete the survey; this was far from ideal. Because most students took only the bare minimum time to complete the survey, response times were most strongly related to the time required to display each visualization method's animation instead of the time required for mental visualization/rotation. In order to improve the quality of the collected data, several changes to the survey were made before it was run again in the fall of 2012: participants were asked to record their sex before taking the survey, timestamp of survey completion was recorded, the instructions available for each visualization method were lengthened and improved, and the survey's general aesthetics were improved. Images from the refined Fall survey are shown in appendix D.

### Fall 2012 Testing:

The refined survey was run with a new group of PY151 students in the fall of 2012. The data from this experimental run is shown in table 3. Note that the additional gathered data (participant sex and time of completion) allows additional comparisons to be made for the data collected.

|                                                                                | Multiple 2D Value<br>(n=30, 22 male) |                       | Multiple 3D Value<br>(n=27, 20 male) |                       | Rotating Axes Value<br>(n=25, 14 male) |                       |
|--------------------------------------------------------------------------------|--------------------------------------|-----------------------|--------------------------------------|-----------------------|----------------------------------------|-----------------------|
|                                                                                | Mean                                 | Standard<br>Deviation | Mean                                 | Standard<br>Deviation | Mean                                   | Standard<br>Deviation |
| Time required<br>for survey<br>completion<br>(seconds)                         |                                      |                       |                                      |                       |                                        |                       |
| Male                                                                           | 82.34                                | 42.20                 | 70.31                                | 24.15                 | 91.06                                  | 41.47                 |
| Female                                                                         | 107.83                               | 47.05                 | 61.86                                | 17.56                 | 86.43                                  | 43.18                 |
| Combined                                                                       | 89.13                                | 44.98                 | 68.12                                | 22.93                 | 89.03                                  | 42.29                 |
| Questions<br>Answered<br>Correctly (Out<br>of 6)                               |                                      |                       |                                      |                       |                                        |                       |
| Male                                                                           | 4.86                                 | 1.52                  | 5.25                                 | 0.94                  | 5.50                                   | 0.73                  |
| Female                                                                         | 5.38                                 | 0.99                  | 4.57                                 | 1.05                  | 5.18                                   | 1.34                  |
| Combined                                                                       | 5.00                                 | 1.41                  | 5.07                                 | 1.02                  | 5.36                                   | 1.05                  |
| Pearson<br>Correlation<br>Between<br>Number<br>Correct and<br>Time<br>Required | 0.42                                 |                       | -0.12                                |                       | 0.36                                   |                       |

Table 3. Results from Fall 2012 survey

The refined survey resulted in an overall increase in both the time required to complete the survey and the number of questions answered correctly. For the multiple two-dimensional axes method, the mean time of completion increased slightly from 86 seconds to 89 seconds; mean number correct increased from 4.625 to 5 questions correct out of 6. For the multiple three-dimensional axes method, the average time of completion increased from 59 seconds to 68 seconds, while the average score increased from 4.862 to 5.07 questions correct out of 6. For the rotating axes method, the average time of completion increased from 80 seconds to 89 seconds, while the average number of correct responses increased from 5.06 to 5.36. Completion time was measured solely as the time required to answer every question; time required for reviewing answers and reading instructions was not included in this measurement. The participant data for the number of correct answers is shown in figure 2 and the data for completion time is shown in figure 3.

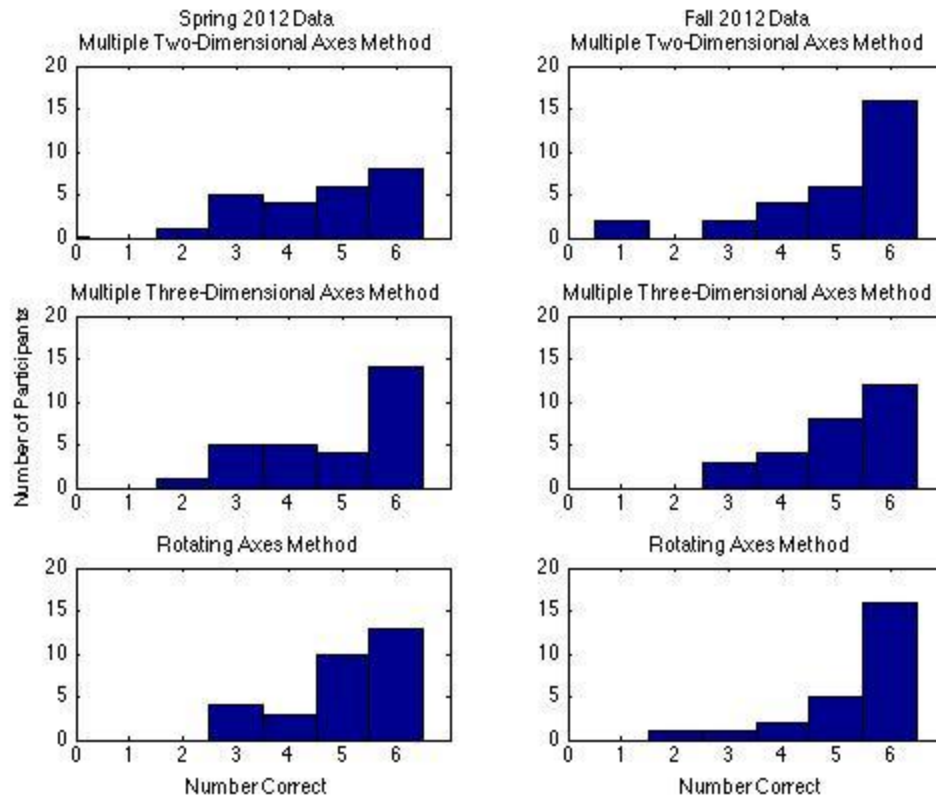


Figure 2: Number of correct answers by semester and visualization method

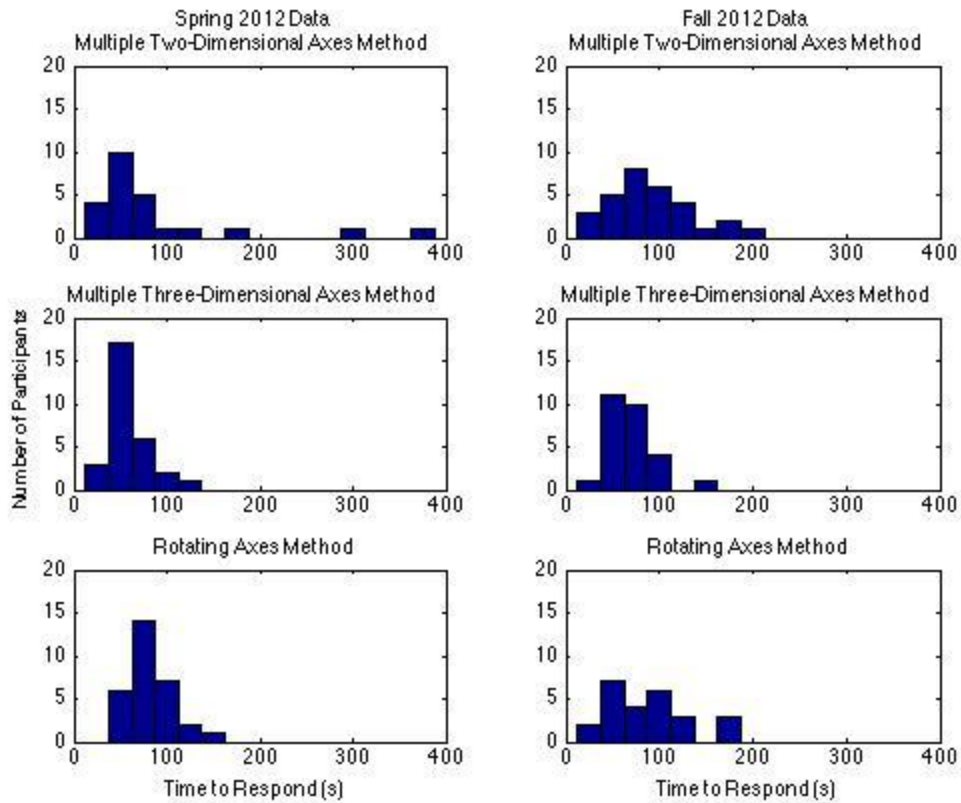


Figure 3: Survey completion times by semester and visualization method

The improved scores and increased response times are likely due to the additional training given for each visualization method. Gender was analyzed as a possible confounding variable, but had little consistent impact on performance- female participants spent more time and performed better on the multiple two-dimensional axes method than male participants, but this trend was reversed for the multiple three-dimensional and rotating axes methods. This result indicates that the more time spent on a question, the better the participant does (on average). However, this conclusion is refuted by the average completion times for each question set: only the rotating axes method shows that the highest average response time was related to the best performance. Thus, it would appear that individual variability in the sampled population played a larger role than response time on the number of questions answered correctly. This conclusion is supported by student's t-tests conducted on the collected response times and number correct. These t-tests are summarized in table 4.

| Comparison                                       | t value (Spring) | Conclusion      | t value (Fall) | Conclusion      |
|--------------------------------------------------|------------------|-----------------|----------------|-----------------|
| between multiple 2d and multiple 3d (# correct)  | -0.66629         | Not significant | -0.22874       | Not significant |
| between multiple 2d and rotating axes (#correct) | -1.36438         | Not significant | -1.08014       | Not significant |



|                                                  |          |                 |          |                 |
|--------------------------------------------------|----------|-----------------|----------|-----------------|
| between multiple 3d and rotating axes (#correct) | -0.66359 | Not significant | -0.99473 | Not significant |
| between multiple 2d and multiple 3d (time)       | 1.500106 | Not significant | 2.254563 | Significant     |
| between multiple 2d and rotating axes (time)     | 0.310176 | Not significant | 1.782818 | Not significant |
| between multiple 3d and rotating axes (time)     | -3.63133 | Significant     | -2.19167 | Significant     |

Table 4: T-test results

These T-tests show that the differences between the number of correct answers for all three conditions were not significant. While the time required for completion of each question set was significant for the multiple three-dimensional comparisons to the other methods, this difference was likely due to the display time required for this method. Half of the three-dimensional figures shown were still images, which do not require an animation to display and thus can be answered more quickly than a question requiring an animation. This would account for the statistically significant difference in completion times between the multiple three-dimensional axes method and the other methods investigated.

Additional statistical analysis done on both the spring and fall data is shown in Appendix E.

## Conclusions:

Based on the data gathered in the Spring and Fall of 2012, no significant differences in the number of questions answered correctly could be found between the three visualization methods explored. Training effects as determined by the instructions available to students had a significant impact on mean questions correctly answered and the time required to do so.

## Discussion:

Visualization of data in more than three dimensions is a difficult problem for human spatial cognition; based on the experiments described above, at most 64% of participants were able to understand three and four-dimensional objects represented using a heuristic-based visualization method. This number could certainly be improved through two main factors: increased student motivation and increased participant training. Students were motivated primarily by course credit, which was contingent on completion of the survey and not performance; most took approximately two to three minutes to complete the survey. The time spent on each question could have been improved by relating performance with reward (specifically the monetary reward offered to a randomly selected student).

Due to IRB requirements about connecting participant data to rewards without a physical copy of a signed informed consent form, we were unable to pay participants based on the number of questions they correctly answered. Instead, we motivated students via a random drawing of two participants who received a fixed sum. Changing this reward scheme may have increased student response time and questions correctly answered. Additional training provided in the fall 2012 survey (but not the spring 2012 survey) increased participant response times and mean questions correctly answered. This implies that additional training and/or practice questions could further improve student responses.

Analysis of the number of correct answers for each method showed no significant differences. Significant differences could be found more readily by increasing the sample size for each method. One possible experiment to do this would involve switching from a between subjects experimental design to a within subjects design (showing every participant all three surveys). The order of survey presentation would be counterbalanced in order to prevent training effect confounds. This change to the experiment would effectively triple the sample sizes for each set, possibly revealing more significant conclusions. This experiment was not implemented due to time constraints, but would be worth investigating in more detail.

## Program Design

### N-Dimensional Graphing Software

The N-Dimensional Graphing software is built around the idea of a variable-length vector containing point information. The user interface is used to open submenus, each of which changes certain parameters of the stored information. The current “universe” of stored points, lines and planes can be saved or loaded from text files. In addition, the software is able to read point information from excel files. The overall block diagram for this software is shown in figure 4.

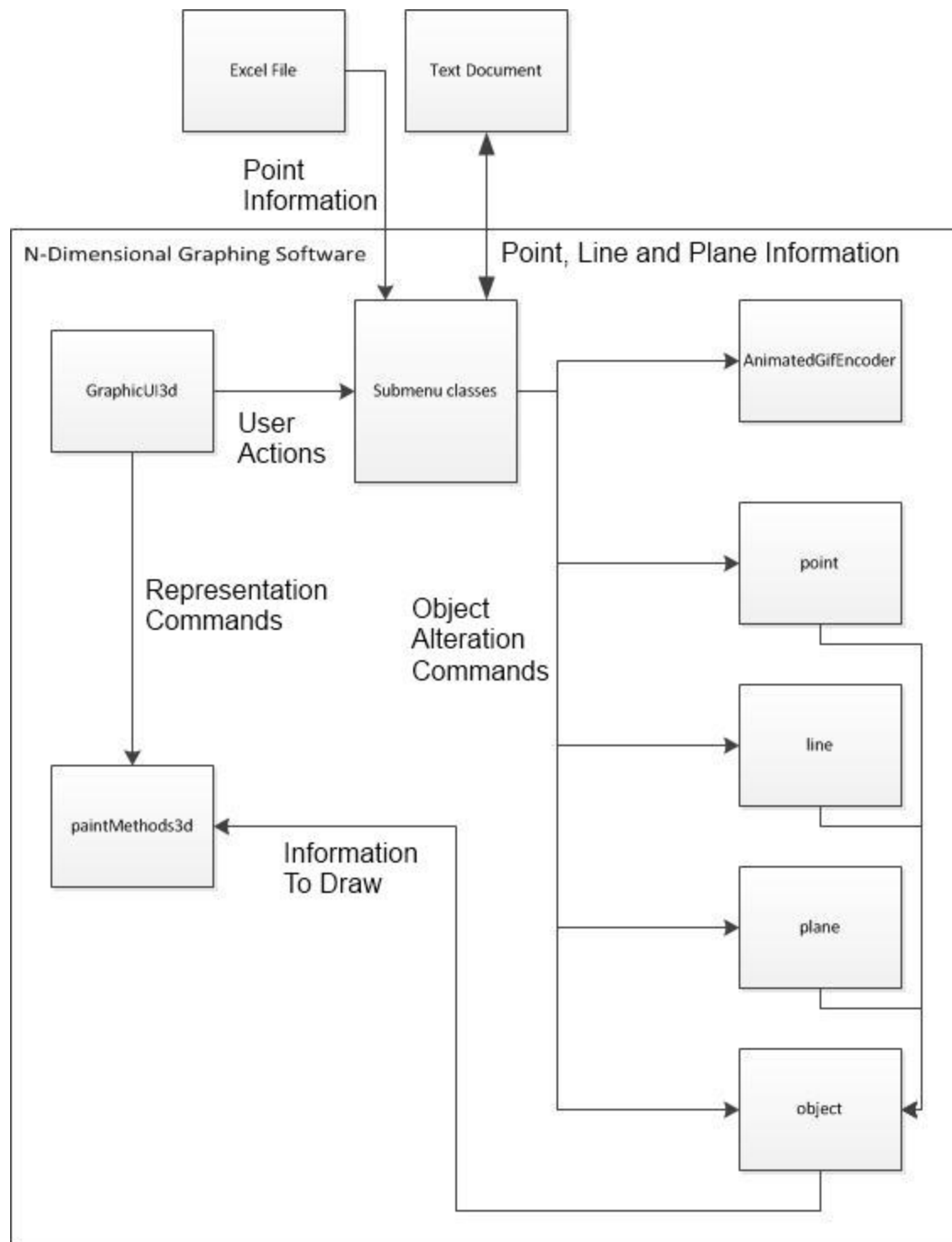


Figure 4: Block Diagram for N-Dimensional Graphing software

Each of the subcomponents in this block diagram is required for the software to function. The GraphicUI3d class generates the user interface (a menu bar at the top of the screen and a canvas to draw images on). The user interface appears as shown in figure 5.

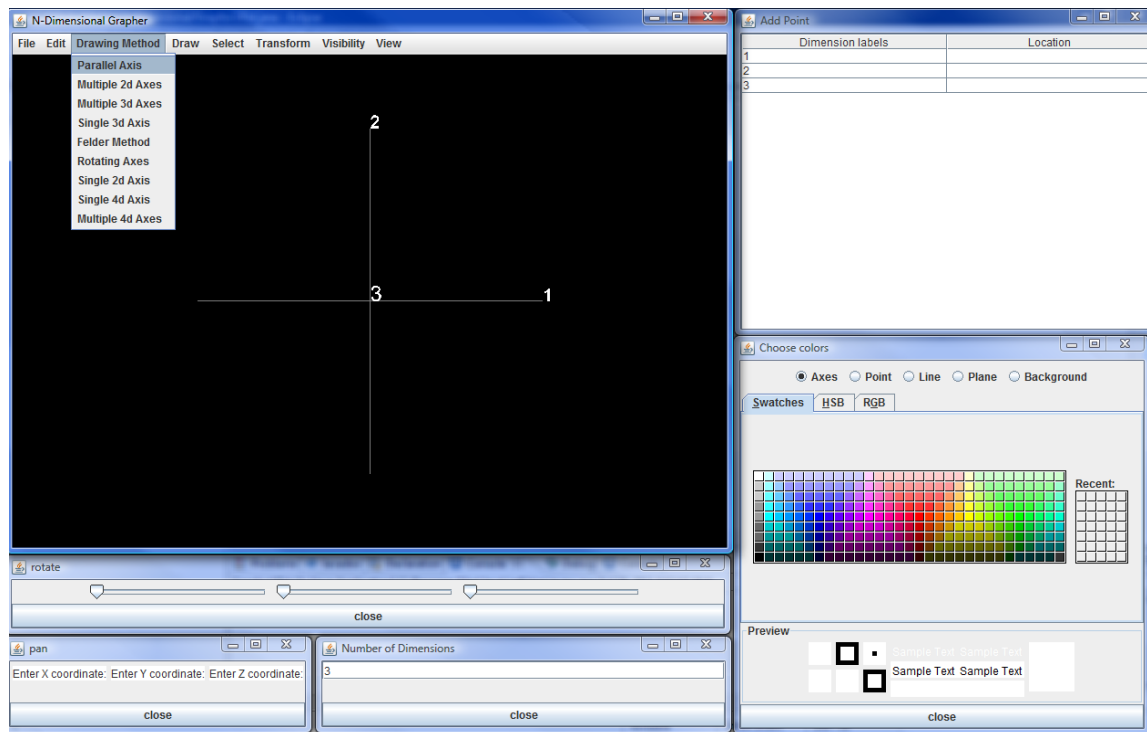


Figure 5: User Interface for n-dimensional graphing software (with multiple submenus open)

When a menu option is clicked, it activates submenu code that allows the user to set parameters of the desired change. The program maintains a single instance of the object class- this serves to store all the point, line and plane information entered. Points are specified as locations in n-dimensional space, where n is specified by the user. Lines and planes are defined by their endpoints- two for lines, three for planes. Based on the user's input, the `paintMethods3d` class draws different representations of the object in the `java3d` canvas space (a three dimensional environment). In addition, the submenus available can open excel files, save to/load from text files and save animations using the `AnimatedGifEncoder` class.

## Online Survey System

The online survey system is a java applet built for use by participants in the survey. In order to conform to proper psychological testing standards, this software was designed to follow these guidelines:

1. Randomized block design- question order can affect participant responses, so each participant sees a different question order
2. Three experimental conditions- participants can only view one condition
3. Data storage- participant responses must be saved to a central location viewable by the experimental team

These specifications were satisfied via apache tomcat webserver, which hosted the survey and servlets used to store relevant data. The overall block diagram for this system is shown in figure 6.

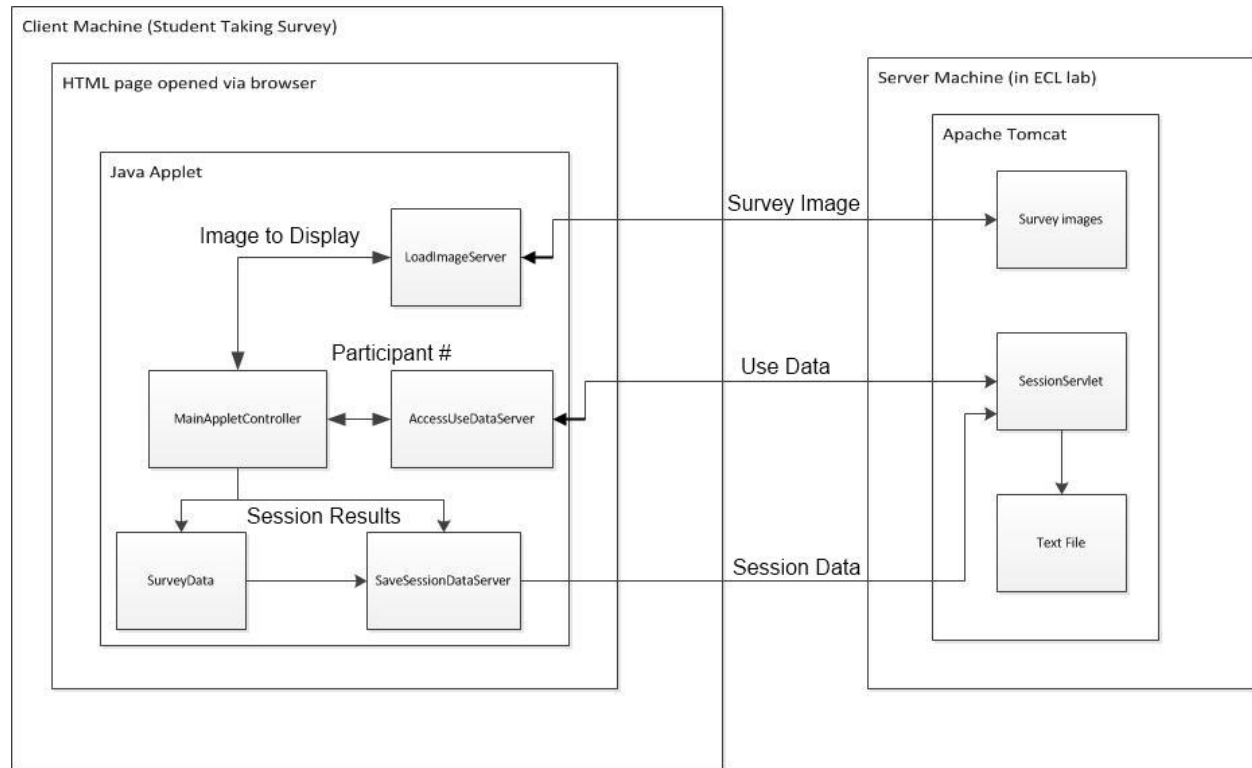


Figure 6: Block Diagram for Online Survey System

As can be seen, there are five main classes that operate on the client (student) side. The **MainAppletController** class acts as the main user interface for the survey- it initializes the survey and controls the individual slides that the user sees. The **LoadImageServer** class is used to load images for each instruction, question and answer slide. The **AccessUseDataServer** class communicates with the server machine to determine which experimental condition/ question order to give the participant. The **SurveyData** class is filled with information from the user's survey answers, including his/her responses and reaction times. The **SaveSessionDataServer** class sends this information to the server so it can be saved.

On the server side, all subclasses are hosted within the Apache Tomcat framework. The active servlet (**SessionServlet**) maintains a count of the number of participants who have taken the survey and saves incoming data as text files on the server. This is used to maintain the randomized block design required for this experiment.

Apache Tomcat allows external computers to see an HTML page at a specified link, which can be accessed anywhere within Clarkson's campus network (due to limitations imposed by the Office of Information Technology it could not be accessed off-campus).

All source code used within this project (for the n-dimensional graphing software, the online survey system and the servlets used to run the survey system) is available at:

<http://people.clarkson.edu/~racettl/HonorsSeniorThesis/>

## Appendices

### Appendix A: Visualization methods explored

Five main methods were originally developed for this project. After pilot testing, only three were chosen for further analysis: the multiple two-dimensional axes method, multiple three-dimensional axes method and rotating axes method. The two methods explored in the pilot study and removed from the later studies were the parallel axes method and the Felder method. These visualization methods are detailed below. Note that these images have been inverted to allow for easier printing; the actual images shown to participants were composed of green lines on a black background with white axes, instead of purple lines on a white background with black axes.

#### Parallel Axis Method:

The parallel axis method involves visualizing each axis as being parallel to all others. For this method, points appear as lines connecting each axis. For example, the point  $(0,-1,0,1)$  in four-dimensional space would appear as shown in figure A1.

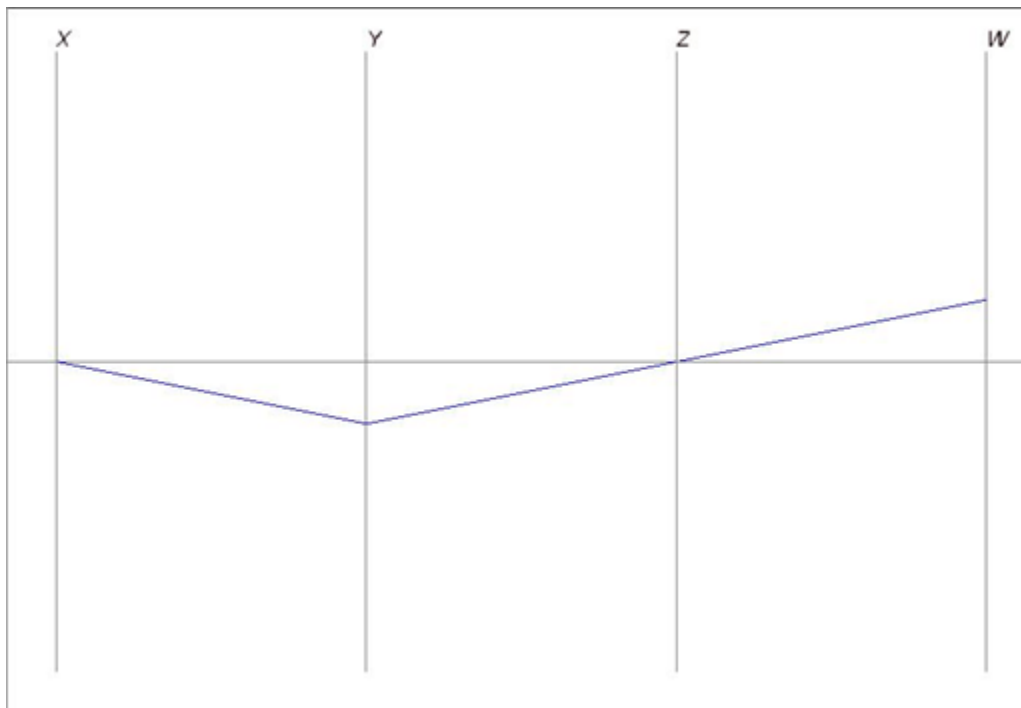
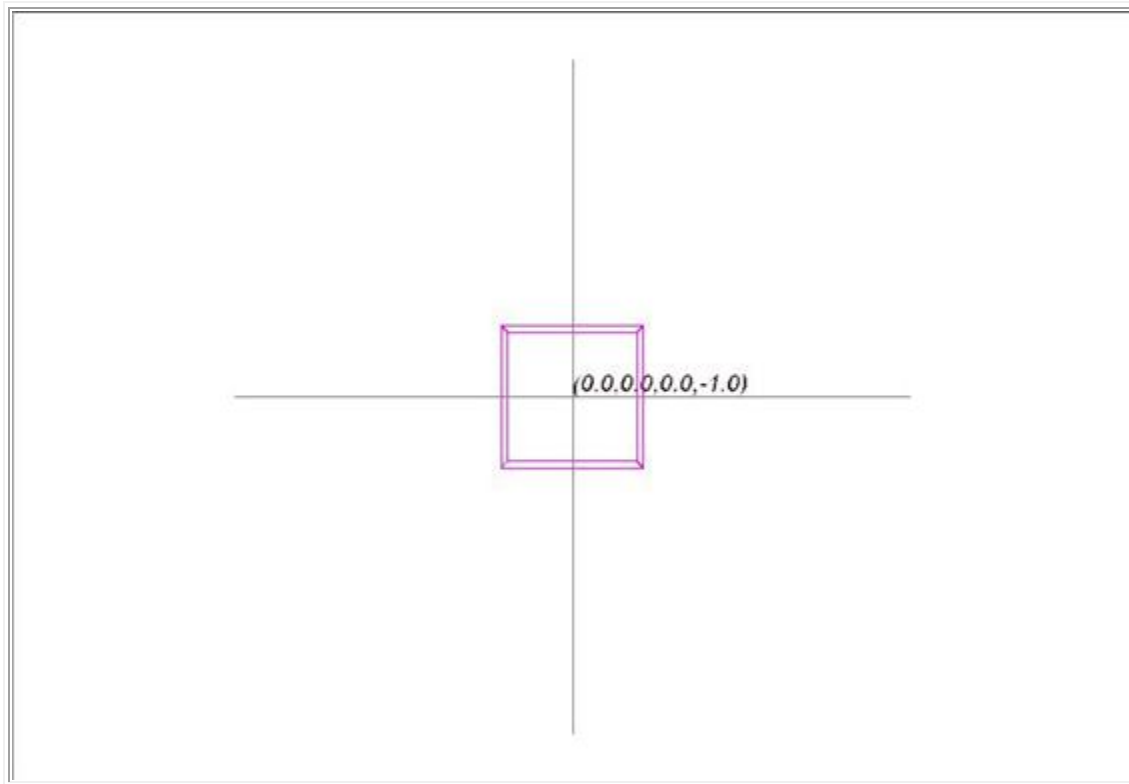


Figure A1:  $(0,-1,0,1)$  in the parallel axis method

**Felder Method:**

The Felder method relies upon "slices" at certain intervals along each dimension higher than the third. This creates a different three-dimensional axis for each of these slices. For reference, the origin location for each set of axes is included in the displayed image. If the currently viewed slice is the  $W=3$  slice (where  $W$  is the fourth dimension), then this origin would read  $(0.0,0.0,0.0,3.0)$ . As an example, a four-dimensional cube is shown in figure A2.





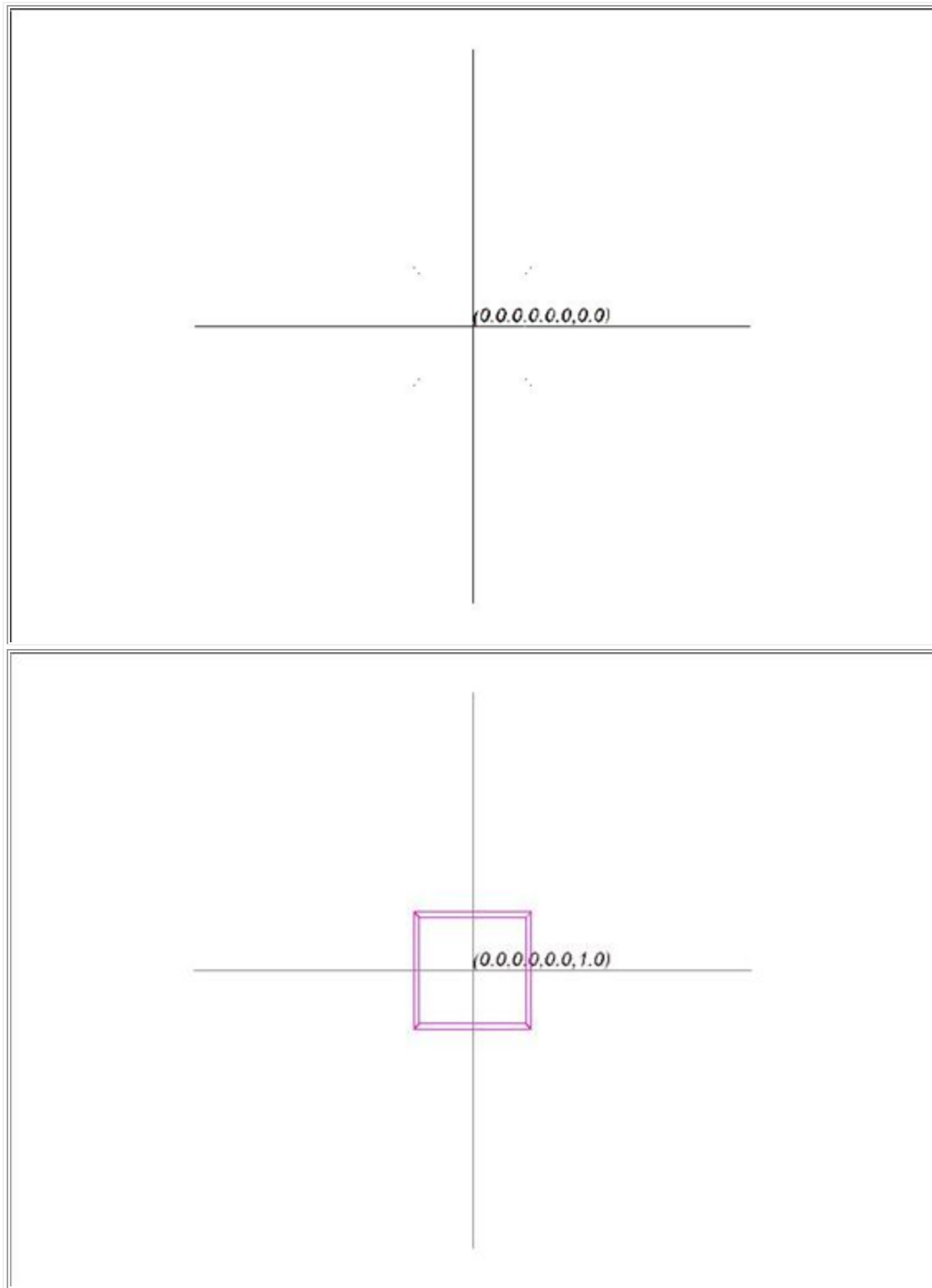


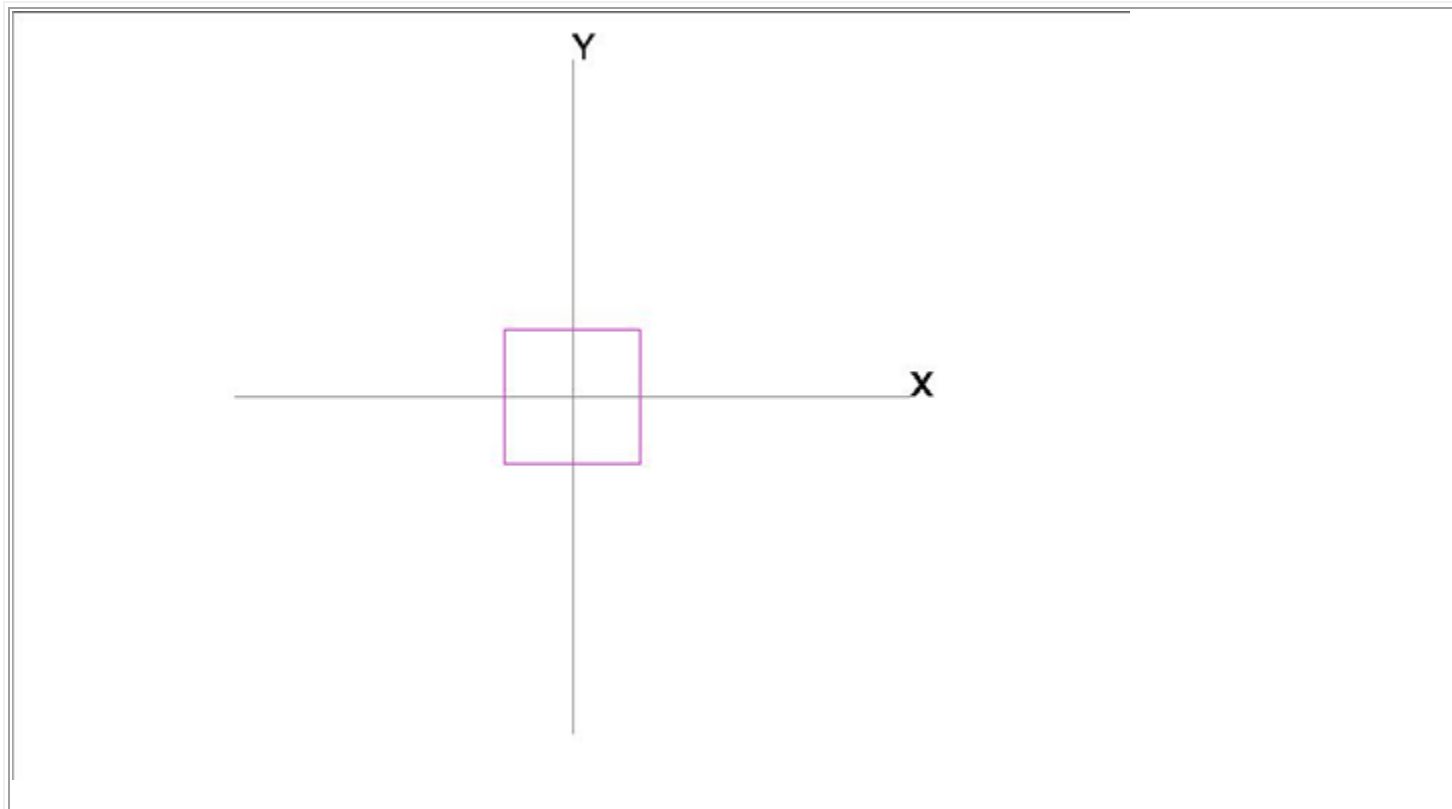
Figure A2: Four-Dimensional Cube in Felder Method Representation

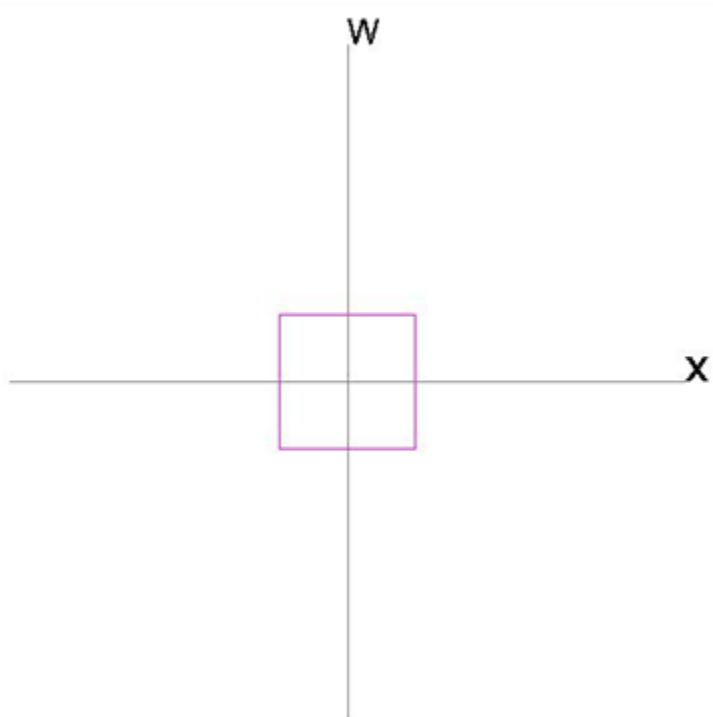
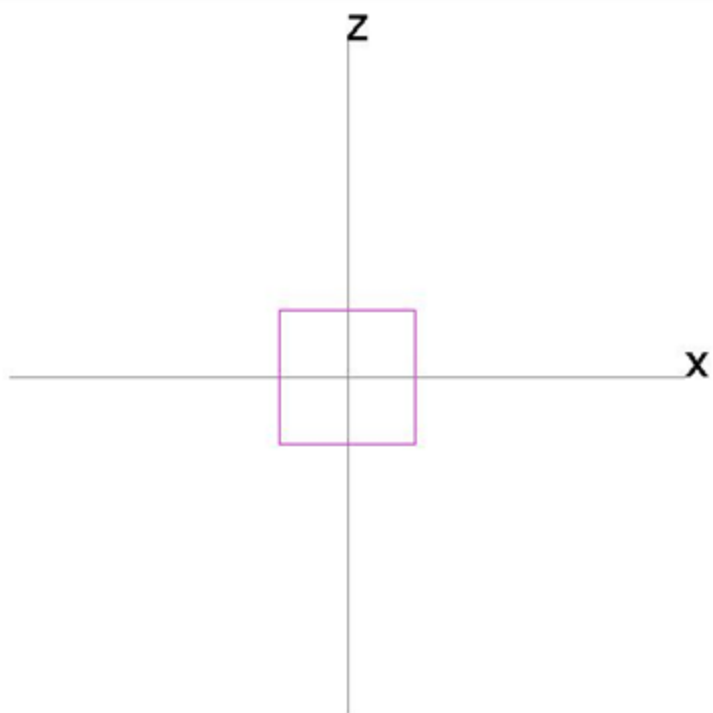
Basically, the Felder method shows two cubes at the slices  $W=-1$  and  $W=1$ . Because this is a four dimensional cube, the vertices of these cubes are connected across the  $W$  dimension. This is why the

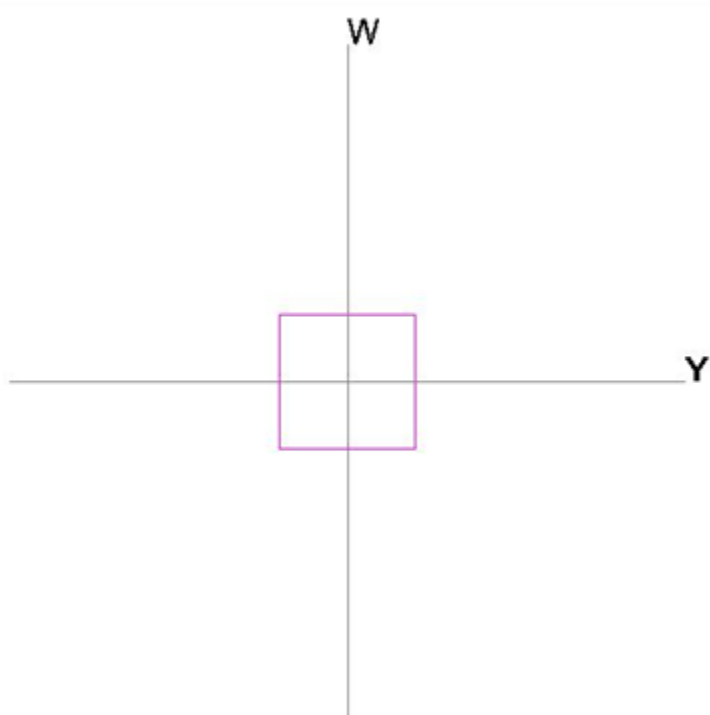
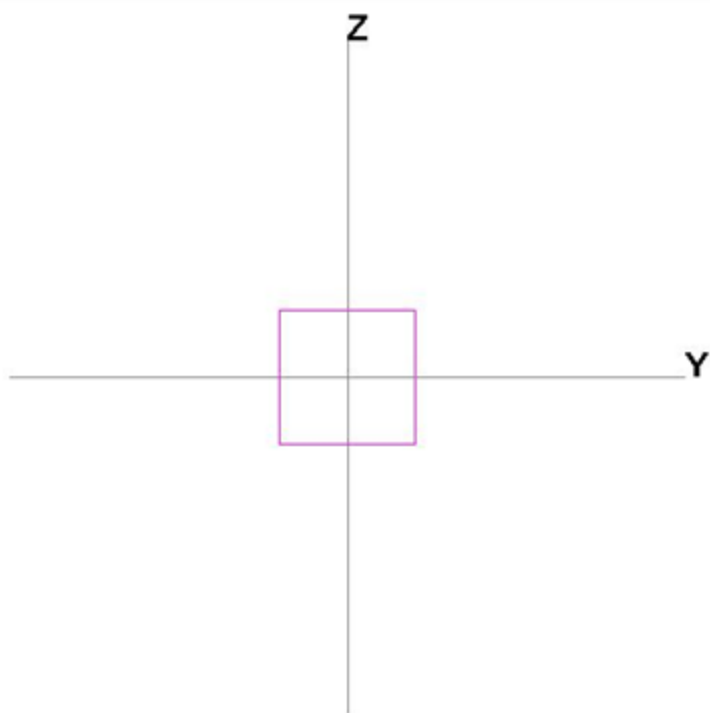
$W=0$  slice shows dots at the vertices of the cube- each dot represents a line that is parallel to the fourth dimension and only passing through the  $W=0$  slice at one point.

**Multiple Two-Dimensional Axes Method:**

The multiple two-dimensional axes method shows a representation of a higher-dimensional object by comparing two dimensions at a time. For example, a single four-dimensional cube would appear as shown in figure A3.







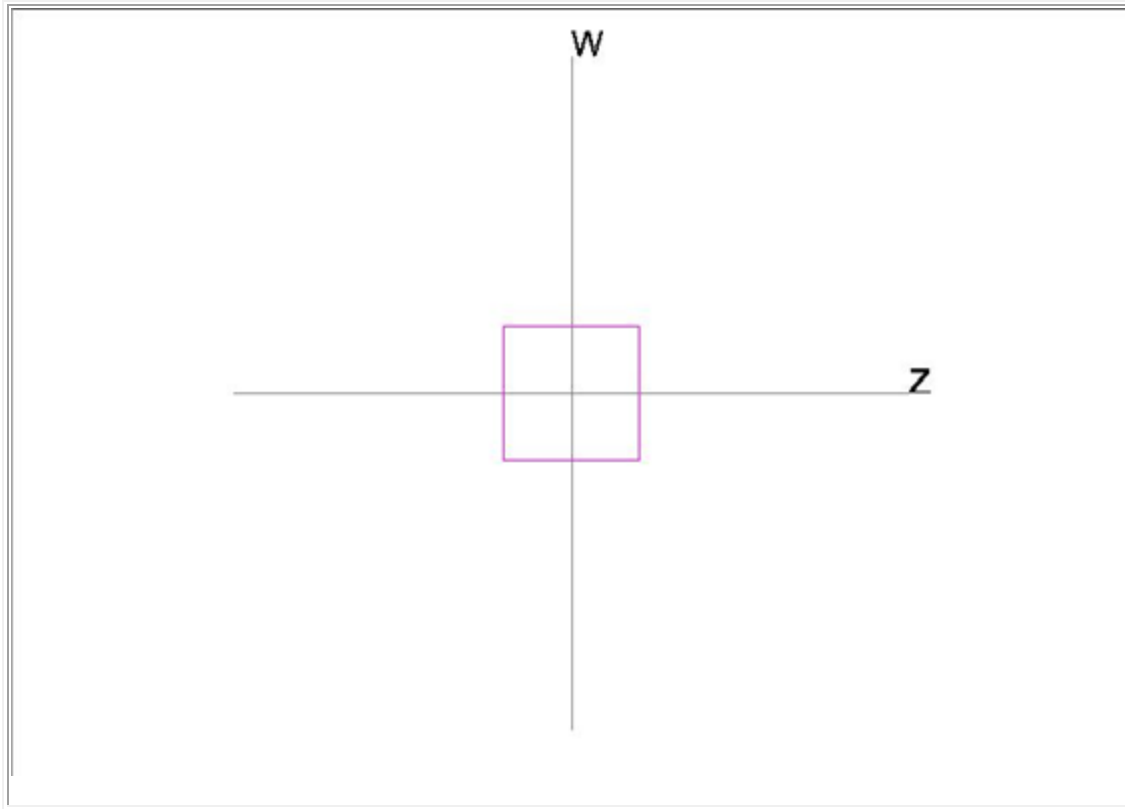
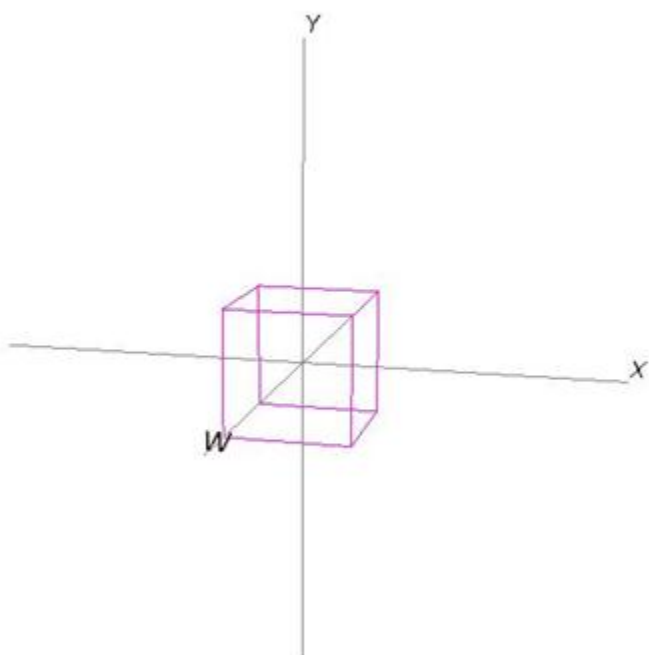
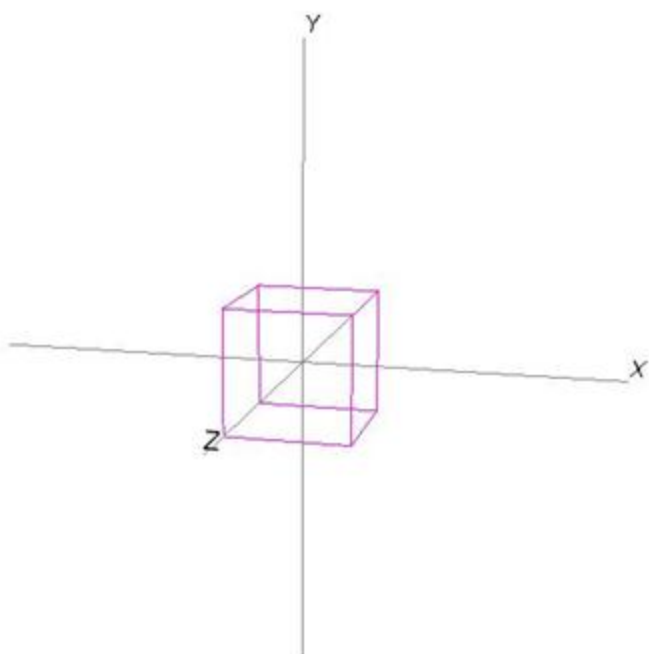


Figure A3: Four-Dimensional Cube in Multiple Two-Dimensional Axes

### Multiple Three-Dimensional Axes Method:

The multiple three-dimensional axes method is similar to the multiple two-dimensional axes method, but displays three dimensions at a time instead of two. A four-dimensional cube displayed using the multiple three-dimensional axes method is shown in figure A4.



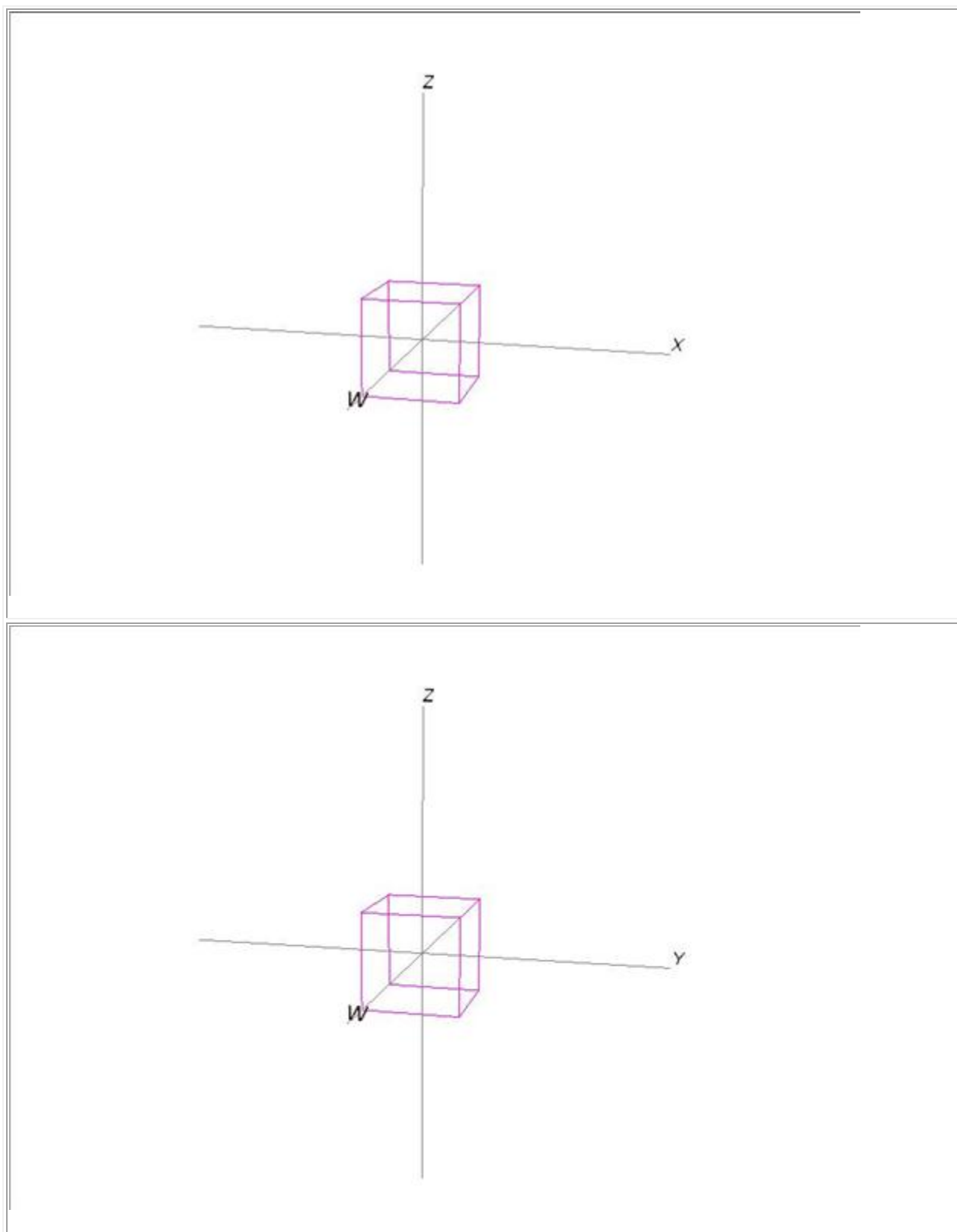
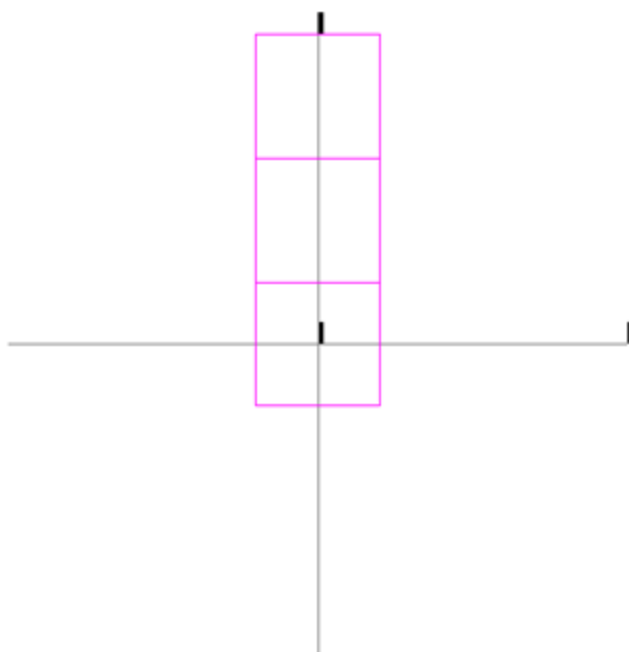
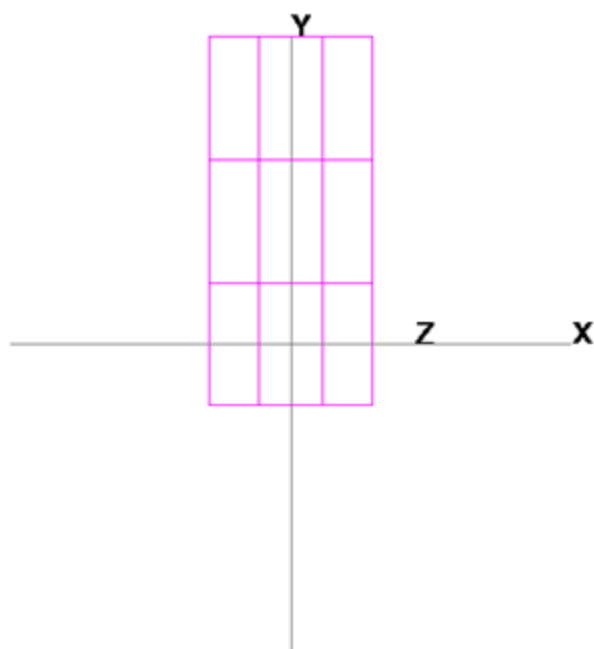


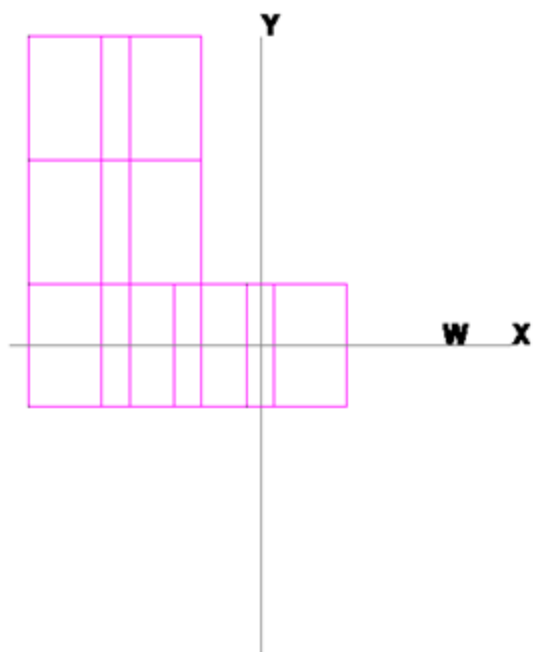
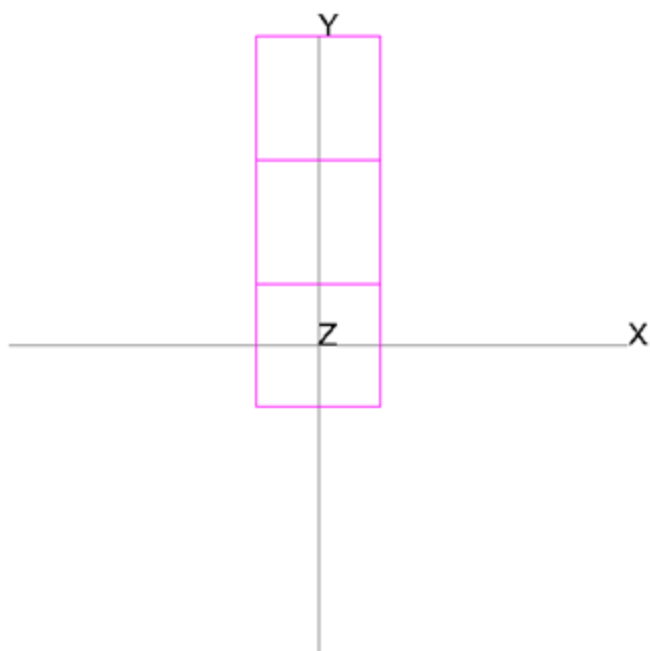
Figure A4: Four-Dimensional cube in multiple three-dimensional axes representation

**Rotating Axes Method:**

The rotating axes method involves using an optical illusion to "see" more than three dimensions. Essentially, the axes continuously rotate back and forth- when viewed perpendicular to the XY axes, the third dimension will change. For only four dimensions, this means that the Z axis is replaced with the W axis or vice-versa. In the actual surveys, this is displayed as an animation. Screen shots from an animation for a four-dimensional figure are shown in figure A5.







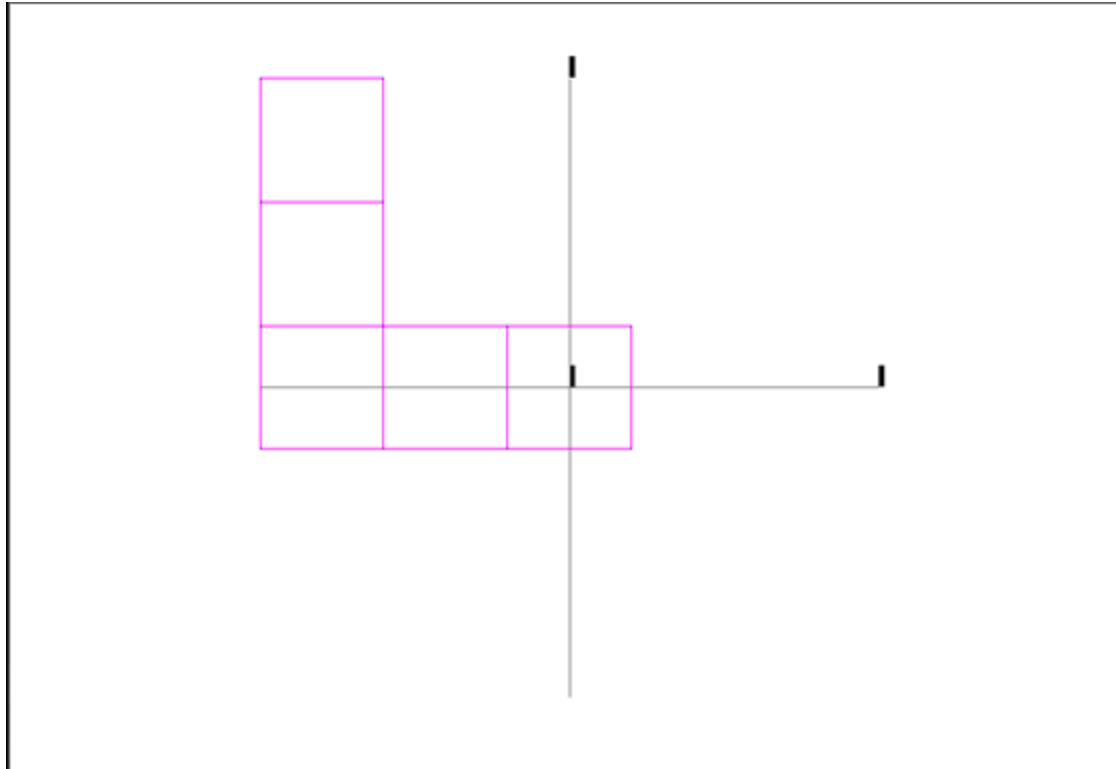


Figure A5: Four-Dimensional Figure in rotating axis representation

## Appendix B: Pilot Surveys

Two Pilot Surveys were completed using responses from members of the ECL (Evolution and Cognition Lab) Team. These were used to refine the experimental methods used for the later surveys. These surveys were made with basic HTML tools, and did not record user information. The first survey consisted of five questions, one for each of the five methods discussed in appendix A. The questions on this pilot survey appeared as shown in figure B1 and answers appeared as shown in figure B2.

## Question 1: Parallel Axis Method

This method involves using parallel axes to display an n-dimensional image. Essentially, every axis ( $x, y, z, w$ , etc.) is placed parallel to the other axes. Points are represented as blue lines between these axes. For example, the point  $(1, 1, 1, 1)$  would be represented as a line connecting the distance 1 on the  $x$  axis with the distance 1 on the  $y, z$ , and  $w$  axes. All axes are scaled the same.

Figure 1: Base Object

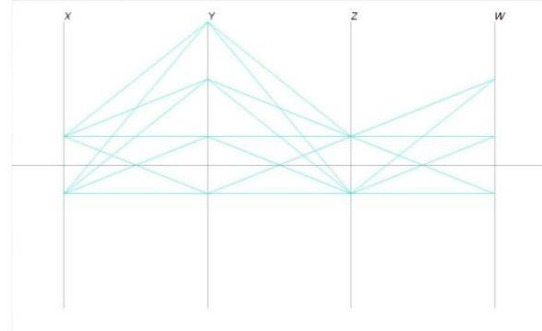
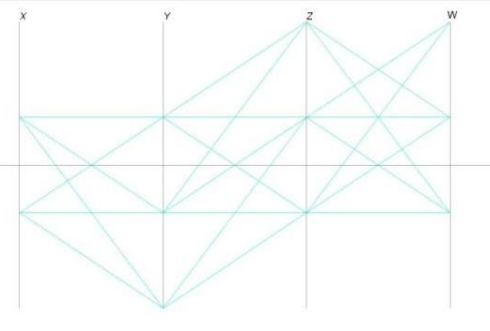


Figure 2: Altered Object



Are these two figures the same object?

- Yes
- No
- Can't Tell

[Answer/Continue](#)

## Figure B1: Pilot Survey 1 Typical Question

## Multiple 2D axes answer

The two objects are in fact the same. If you look closely at the Y-W axes set, the two objects are different, but one could be made from the other by a rotation about the Y-axis.

Answer: Yes

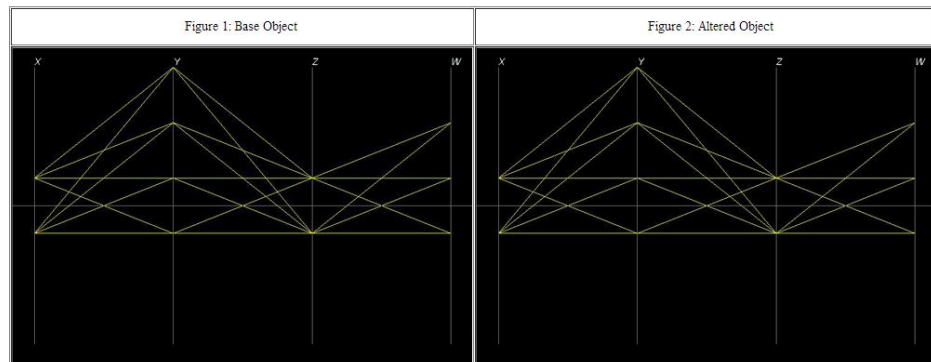
[Next Question](#)

## Figure B2: Pilot Survey 1 Typical Answer

The first pilot survey was expanded to allow for more meaningful comparisons between the visualization methods. The second pilot survey consisted of 30 questions, six for each of the five methods. Each question appeared as shown in figure B3 and answers appeared as shown in figure B4.

Parallel Axis Method, Question 2:

Are these two images representations of the same object?



Yes

No

Can't Tell

[Answer](#)

Figure B3: Pilot Survey 2 Typical Question

Answer: Yes

The two images are the same- close inspection shows that the two images are identical. Note, however, that rotations of the same object will not always produce an identical parallel axis representation.

[Question 3](#)

Figure B4: Pilot Survey 2 Typical Answer

## Appendix C: Survey slides (spring 2012)

The first version of the experimental survey was run in April 2012. This survey gave different question orders and experimental conditions to different students. All students began the survey by viewing an informed consent introduction slide (figure C1) and clicking a button labeled “I have read and accepted this agreement”. Afterwards, students were partitioned into groups based on login order. Each group saw a different instruction slide; underneath this slide were text fields for participant data. Instruction slides appeared as shown in figure C2. After all participant data was collected, the participant saw 6 questions (3 consecutive questions in four-dimensional space and 3 consecutive questions in three-dimensional space). Question order was counterbalanced in both order of the 3D and 4D question sets and question order within these sets, meaning that no two participants saw the same question order. After the first and fourth questions (the first 3D question and first 4D question), the participant was shown an answer slide with an explanation of the preceding question. Questions appeared as shown in figure C3 and answers appeared as

shown in figure C4. After the survey was over, participant data was saved and participants were shown the ending slide in figure C5.

PROJECT TITLE: Visualizing Data in More Than Three Dimensions.

PRINCIPAL INVESTIGATOR: Dr. Andreas Wilke, Evolution and Cognition Lab, Department of Psychology, Clarkson University

PROJECT DESCRIPTION: The purpose of this study is to better understand how individuals comprehend information in hyper dimensional space. With your participation in this web-based questionnaire survey, you can experience how psychologists study internalized object representation in novel environments and obtain a better understanding of how psychological surveys are constructed. This web-based study will take about 1 hour. Participants will receive 1 PY151 research credit.

**IMPORTANT GENERAL INFORMATION:**

Participation in this research is entirely voluntary and you can quit filling out the survey at any time. Deciding not to take part or to stop being part of this research will result in no fine or loss of benefits to which you otherwise have a right. There are only minimal risks associated with participating in this survey and your responses will remain confidential. Identifying information that you may provide in order to get credit for your participation will be kept strictly confidential. No one will have access to your data other than the investigators who are directly responsible for this survey. All data will be stored on secure servers and will be combined for analysis.

Questions about the survey can be directed to Dr. Andreas Wilke, Psychology Department (awilke@clarkson.edu, x7023), who is the principal investigator in this study. If you have any questions concerning your rights as a research subject, if you wish to report any harm, injury, risk, or other concern, or if you felt pressured to participate in this study please contact Dr. Johndan Johnson-Eilola, Associate Chair of the Clarkson University Institutional Review Board (IRB) for human subjects research (johndan@clarkson.edu, x6488). You may print this page for future reference. Thank you.

By clicking the button below, you are indicating that you have read and understand the above information, that you agree to participate in this survey, and that your age is at least 18 years (or 16 years with parental permission). If you do not agree please exit the survey now.

I have read and accepted this agreement

Figure C1: IRB informed consent form slide

Welcome to the Survey of Visualizing Data in More Than Three Dimensions.

In this study, we are interested in testing the effectiveness of several methods of data representation in multiple dimensions. The survey is composed of six questions, all of which relate to the same method (detailed below). Half of the questions will utilize three-dimensional space, and the other half will utilize four-dimensional space.

For each question, you will be presented with two animated figures. The figure on the left will represent the "base object," and the figure on the right will be an altered form of the base. The altered figure may simply be a rotation of the base object, or it may be altered such that it is now an altogether different object. Your task will be to determine whether or not the two figures are representations of the same object. You will be given an answer after the first question in every dimensional set. Note that some of the figures are animations, and may take several seconds to load. Please take this survey on an Ethernet (wired) internet connection in order to correctly display every image.

Your task involves the multiple three-dimensional axes method. In this method, three-dimensional projections of the figure will be shown at 1 second intervals. For a three-dimensional figure, there will be 1 projection (XYZ). This will be shown as a still image. For a four-dimensional figure, there will be 4 sets of projections (XYZ, XYW, XZW, YZW). This animation will repeat indefinitely.

Once the survey has closed, five participants will be selected at random to receive \$5 for every correct answer that they have selected (i.e., for a possible total of \$30 in winnings).

Please fill in your personal information below and then click 'Next' to begin the survey.

Enter your name:  Enter your PY151 professor's name:  Enter your student ID:

Figure C2: Typical Introduction Slide

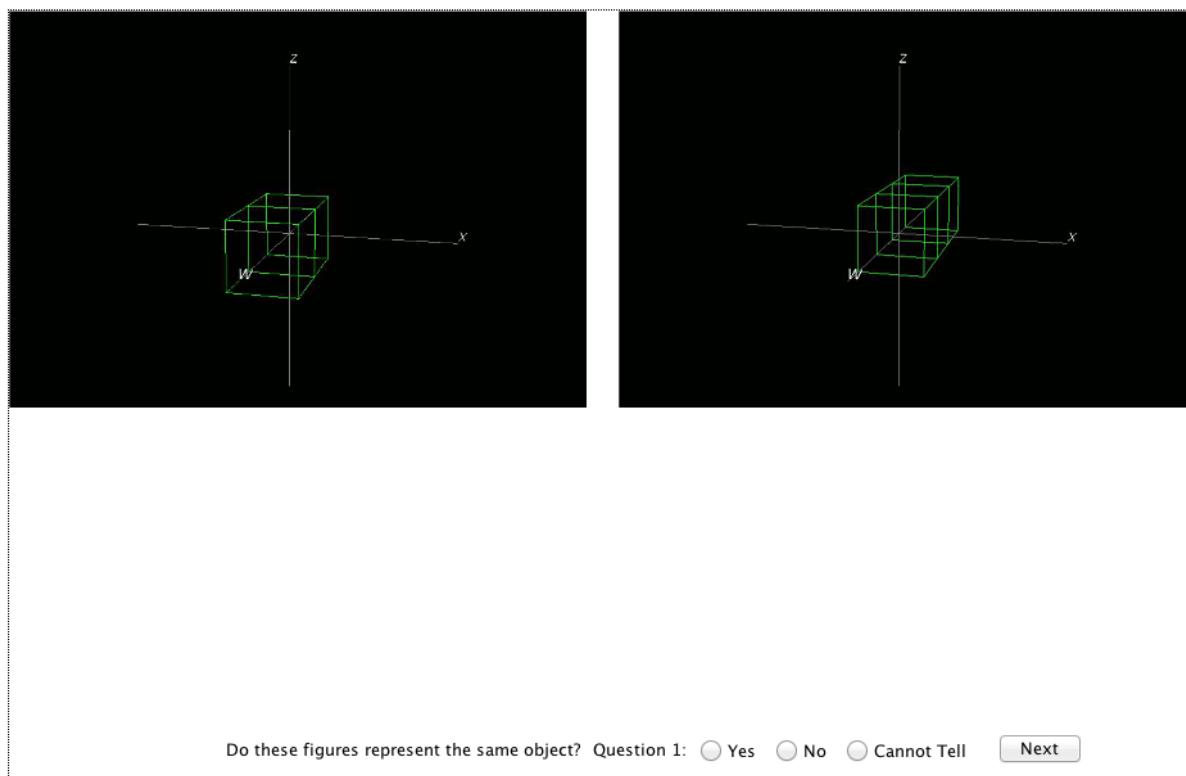


Figure C3: Typical Question Slide



---

The two objects shown are the same. Note that both have three cubes in a line, with a fourth cube perpendicular to this line in four-dimensional space.

Next

Figure C4: Typical Answer Slide

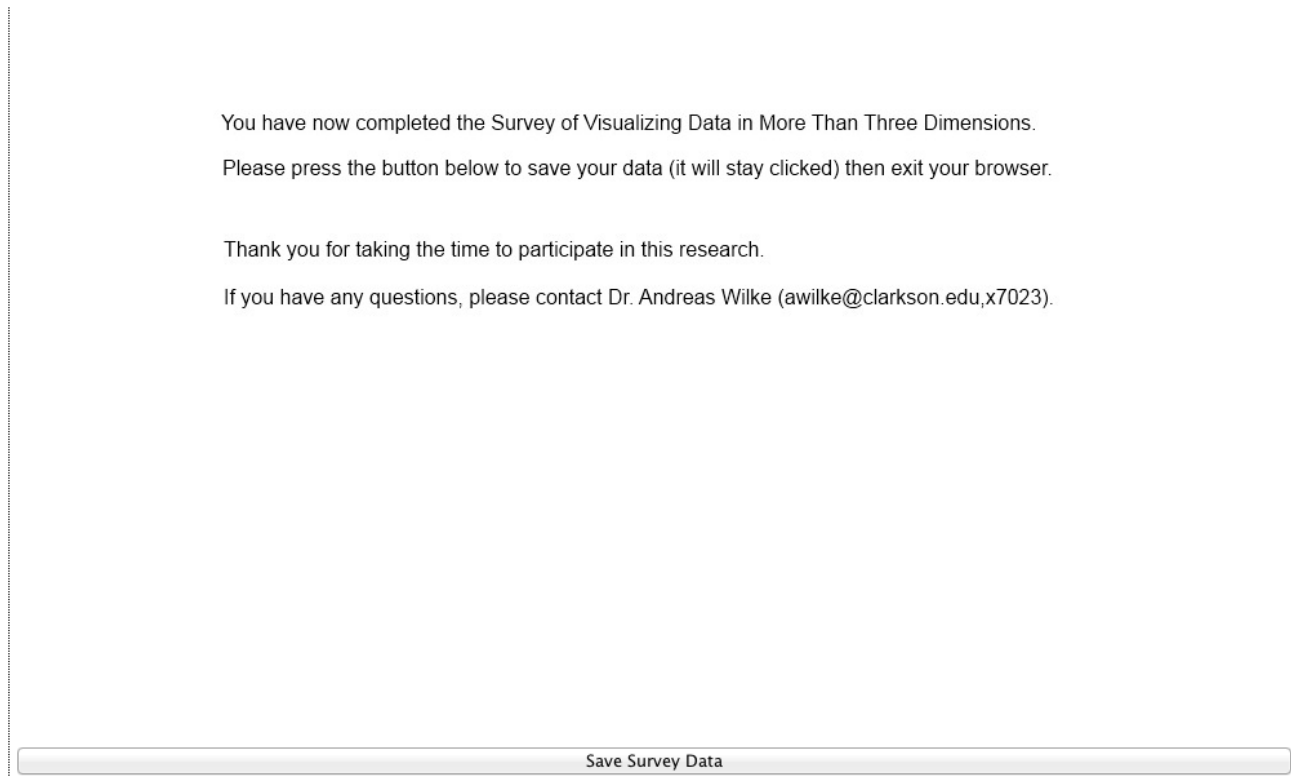


Figure C5: Ending Slide

## Appendix D: Survey slides (fall 2012)

The revised survey run in September- October 2012 had several improvements. These included:

- Extended instruction/introduction slides
- Images shown on answer slides
- Sex selection radio buttons
- Timestamp of survey completion stored with data

Survey order was essentially the same as the Spring 2012 survey, with two exceptions. First, the participants were shown answer slides after every question slide. Second, instructions were shown as four consecutive slides for each condition. Images from a typical survey are shown in figures D1 through D8. Note that images D1 and D8 were seen by every participant, while images D2 through D7 varied based on the experimental condition.

PROJECT TITLE: Visualizing Data in More Than Three Dimensions

PRINCIPAL INVESTIGATOR: Dr. Andreas Wilke, Evolution and Cognition Lab, Department of Psychology, Clarkson University

PROJECT DESCRIPTION: The purpose of this study is to better understand how individuals visualize/conceptualize information shown in more than three dimensions. Participation in this experiment will provide you with experience in how psychologists study internalized object representation in novel environments, as well as general experience with how psychological surveys are constructed. This web-based study will take about 20 minutes. Participants will receive 1 PY151 research credit.

**IMPORTANT GENERAL INFORMATION:**

Participation in this research is entirely voluntary and you can quit filling out the survey at any time. Deciding not to take part or to stop being part of this research will result in no fine or loss of benefits to which you otherwise have a right. There are no more than minimal risks associated with participating in this survey and your responses will remain confidential. No one will have access to your data other than the investigators who are directly responsible for this survey. All data will be stored on secure servers and will be combined for analysis.

Questions about the survey can be directed to Dr. Andreas Wilke, Psychology Department (awilke@clarkson.edu, x7023), who is the principal investigator in this study. If you have any questions concerning your rights as a research subject, if you wish to report any harm, injury, risk or other concern, or if you felt pressured to participate in this study please contact Dr. Johndan Johnson-Eilola, Associate Chair of the Clarkson University Institutional Review Board (IRB) for human subjects research (johndan@clarkson.edu, x6488). You may print this page for future reference. Thank you.

By clicking the box below, you are indicating that you have read and understood the above information, that you agree to participate in this survey, and that your age is 16 years or older. If you do not agree please exit the survey now. If you are under 18 years old, clicking this box also attests that you have a completed parental permission form to participate in this study.

I have read and accepted this agreement

Figure D1: Standard IRB approval slide

Instructions Page 1 of 4

Welcome to the Survey of Visualizing Data in More Than Three Dimensions.

In this study, we are interested in testing the effectiveness of several methods of data representation in multiple dimensions. The survey is composed of six questions, all of which relate to the same method.

For each question, you will be presented with two animated figures. The figure on the left will represent the "base object," and the figure on the right will be an altered form of the base. The altered figure may simply be a rotation of the base object, or it may be altered such that it will be an altogether different object. All figure comparisons will be in the same number of dimensions. Your task will be to determine whether or not the two figures are representations of the same object.

Note that some of the figures are animations, and may take several seconds to load. Please take this survey on an Ethernet (wired) internet connection in order to correctly display every image.

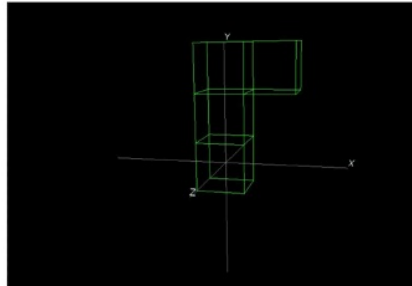
Next

Figure D2: Typical General instructions slide

Instructions Page 2 of 4

Your task involves the **multiple three-dimensional axes** method. Half of the figures will be shown in three-dimensional space, and the other half will be shown in four-dimensional space. For a three-dimensional figure, there will be 1 projection (XYZ). This will be shown as a still image. A still image of a three-dimensional figure is shown below.

(XYZ)



Back

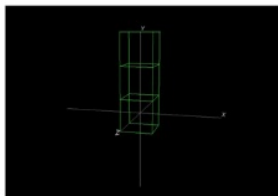
Next

Figure D3: Typical three-dimensional example slide

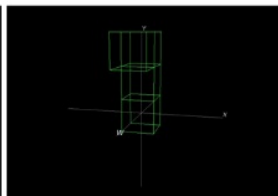
Instructions Page 3 of 4

A four-dimensional figure will be shown as 4 sets of projections (XYZ, XYW, XZW, and YZW) that will alternate indefinitely at 1 second intervals. Still images from a four-dimensional animation are shown below.

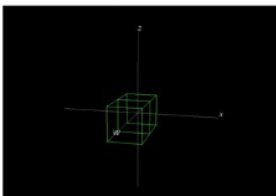
(XYZ)



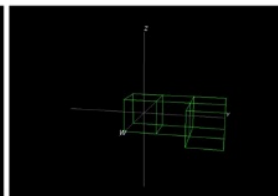
(XYW)



(XZW)



(YZW)



Back

Next

Figure D4: Typical four-dimensional example slide

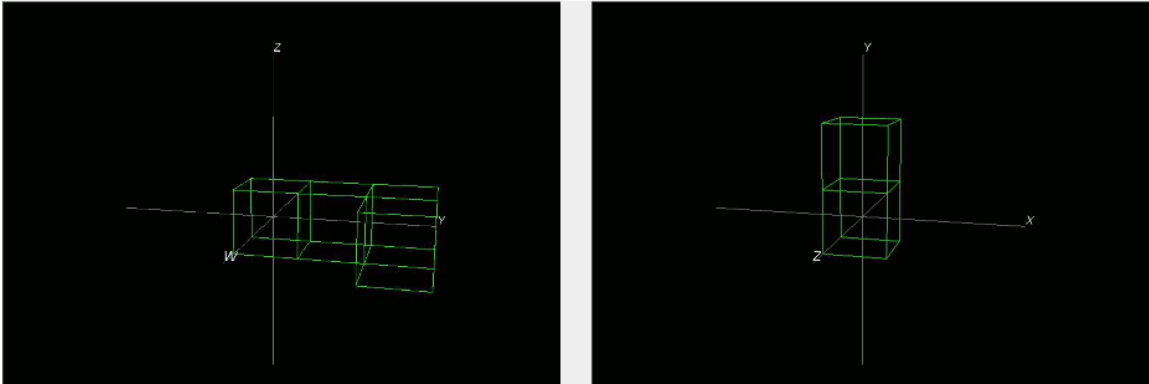
Instructions Page 4 of 4

Once the survey has closed, two participants will be selected at random to receive \$40 for their participation. All participants will receive their PY151 course credit after the survey has closed on Monday, October 8th. You may only take the survey once.

Please fill in your personal information below and then click 'Next' to begin the survey.

Full Name:  PY151 Professor's Name:  Student ID:  ☐ Male ☐ Female

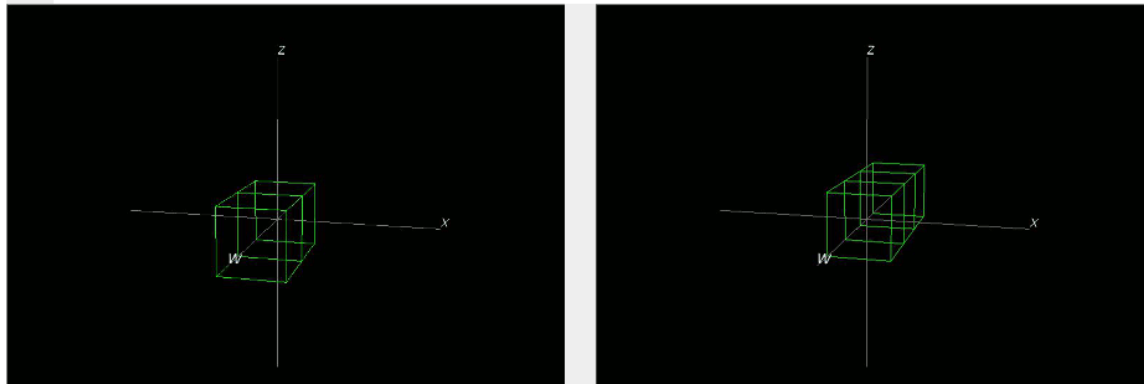
Figure D5: Standard data collection slide



Question 1: Do these figures represent the same object? ☐ Yes ☐ No ☐ Cannot Tell

Figure D6: Typical question slide

The two objects shown are the same. Note that both have three cubes in a line, with a fourth cube perpendicular to this line in four-dimensional space.



Next

Figure D7: Typical answer slide

You have now completed the Survey of Visualizing Data in More Than Three Dimensions.  
Please press the button below to save your data then exit your browser.

Thank you for taking the time to participate in this research.  
If you have any questions, please contact Dr. Andreas Wilke ([awilke@clarkson.edu](mailto:awilke@clarkson.edu), x7023).

Save Survey Data

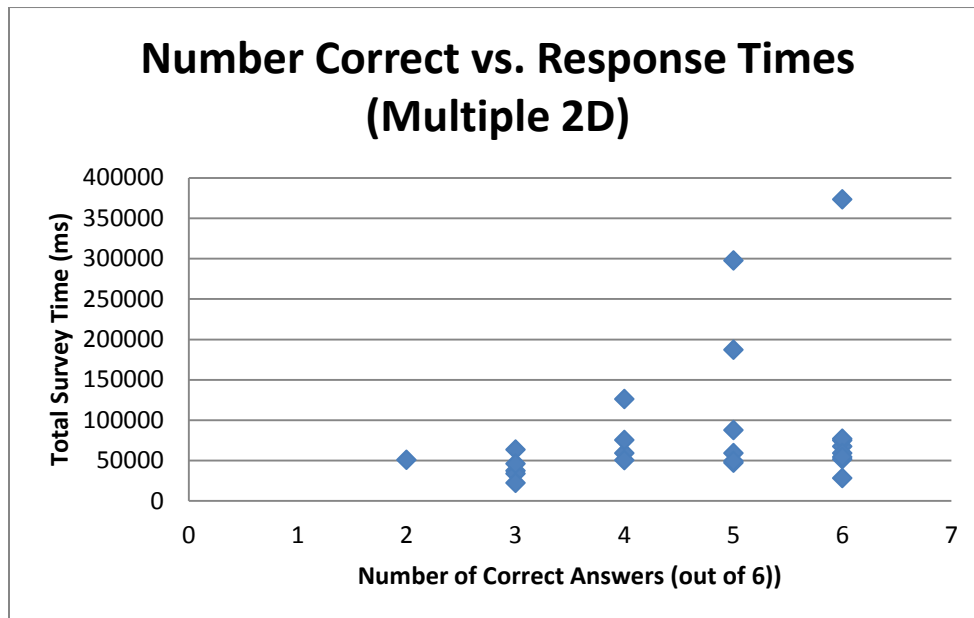
Figure D8: Standard survey ending slide

## Appendix E: Additional Survey Statistics

Additional analysis was done on the collected survey data after completion. This analysis did not directly relate to the working hypothesis, but was useful for understanding and quantifying the trends seen in participant responses.

### Spring 2012:

The number of correct answers as compared to the time required to complete the entire survey is shown for all three conditions in figure E1.



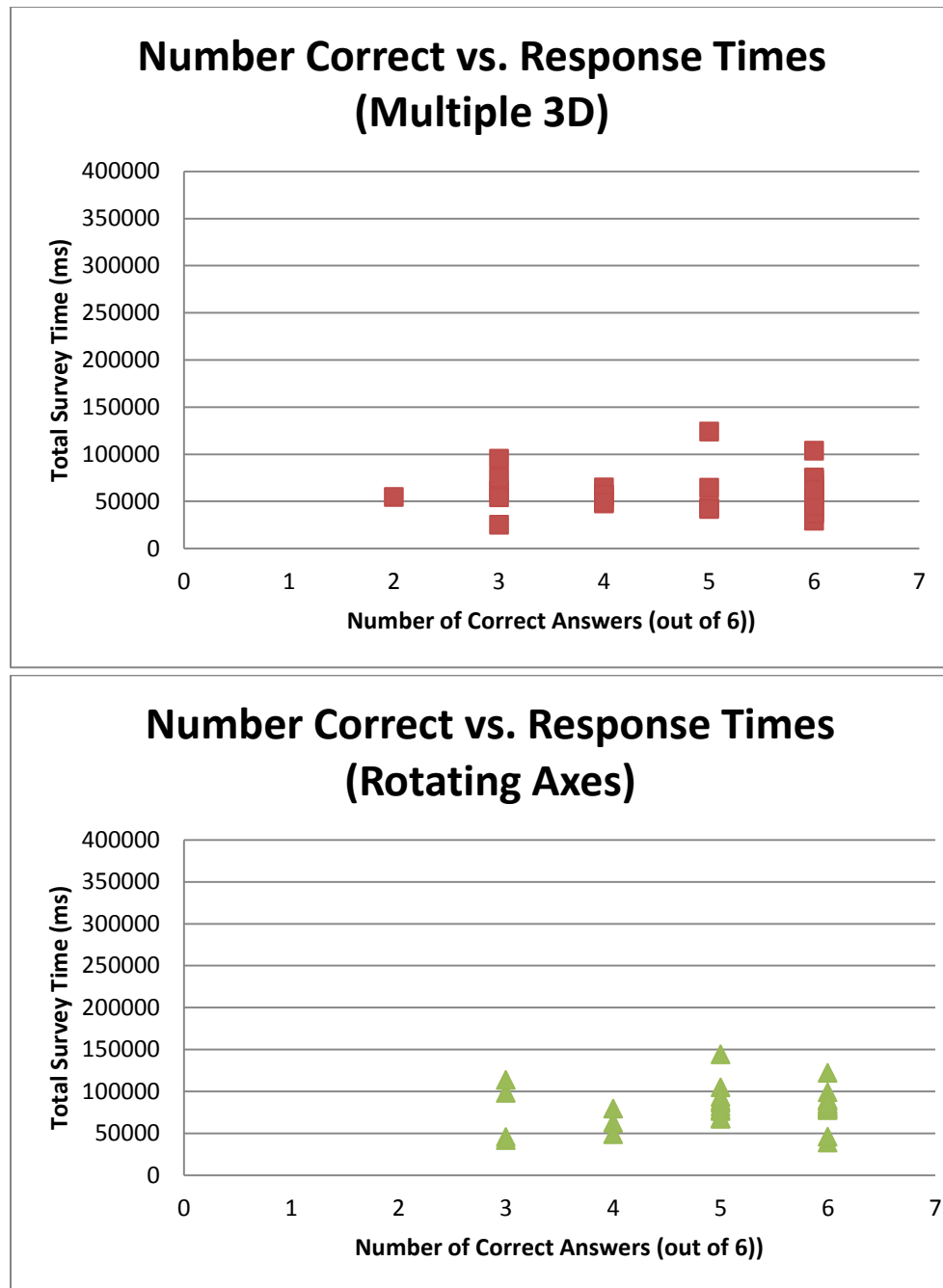


Figure E1: Comparison of number of correct answers to response time for Spring 2012

The average response time for each possible number of correct answers is shown in table E1.

| Measurement                         | Multiple 2D | Multiple 3D | Rotating Axes |
|-------------------------------------|-------------|-------------|---------------|
| Average time (0 correct) in seconds | No Data     | No Data     | No Data       |
| Average time (1 correct) in seconds | No Data     | No Data     | No Data       |
| Average time (2                     | 50.481      | 54.59       | No Data       |

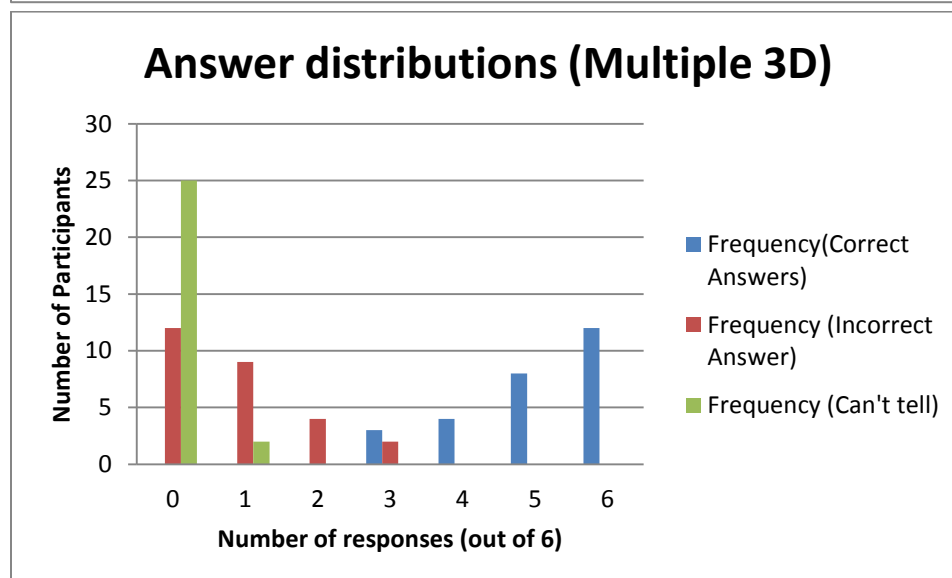
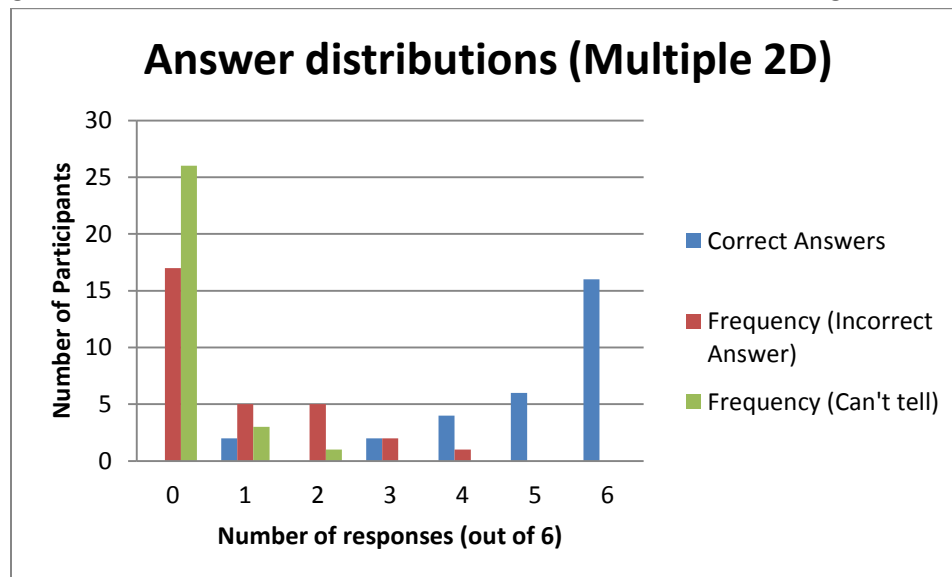


|                                     |          |          |          |
|-------------------------------------|----------|----------|----------|
| correct) in seconds                 |          |          |          |
| Average time (3 correct) in seconds | 40.4878  | 62.2584  | 74.59625 |
| Average time (4 correct) in seconds | 77.564   | 54.775   | 63.42967 |
| Average time (5 correct) in seconds | 121.1545 | 68.3255  | 88.7663  |
| Average time (6 correct) in seconds | 98.08063 | 58.94221 | 80.91346 |

Table E1: Average response times for each number of correct answers

**Fall 2012:**

Extensions to the survey software during the fall survey allowed several additional statistics to be generated. The answer distributions for each method are shown in figure E2.



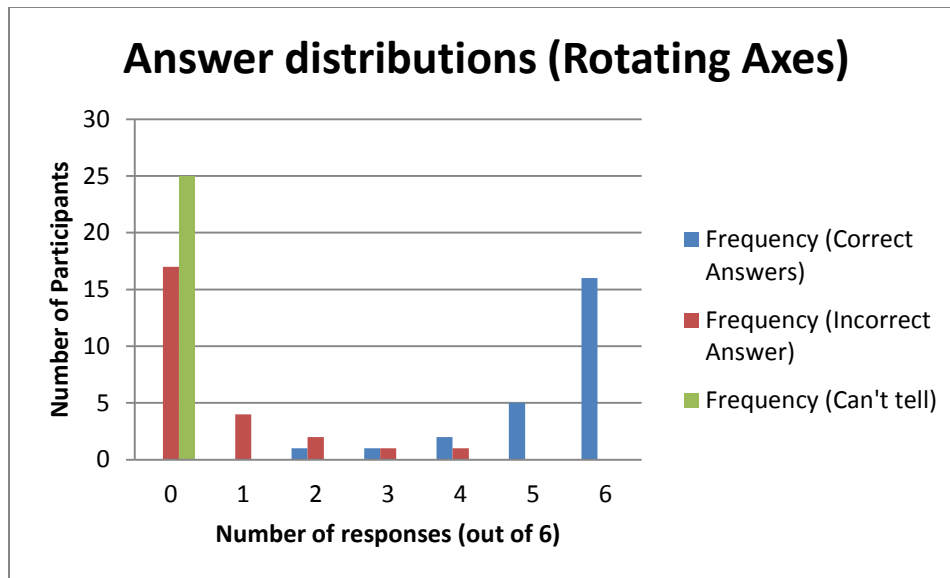
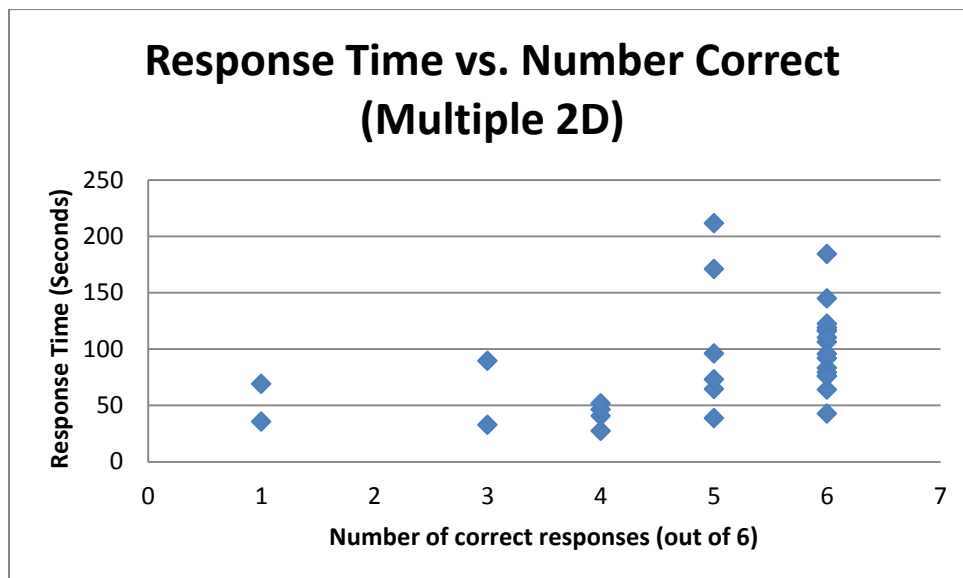


Figure E2: Answer Distributions for Fall 2012 data

The response time as compared to the number of correct answers for each condition is shown in figure E3.



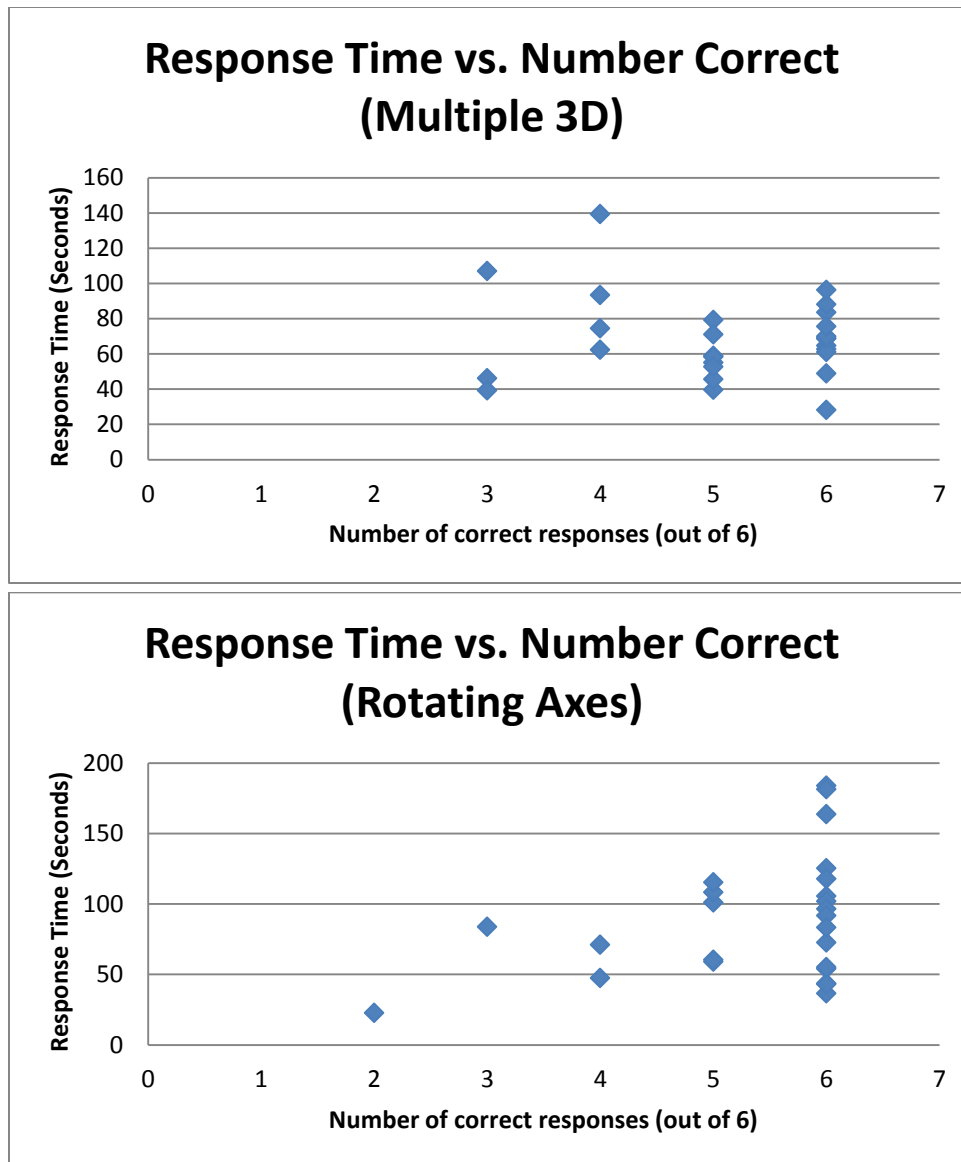
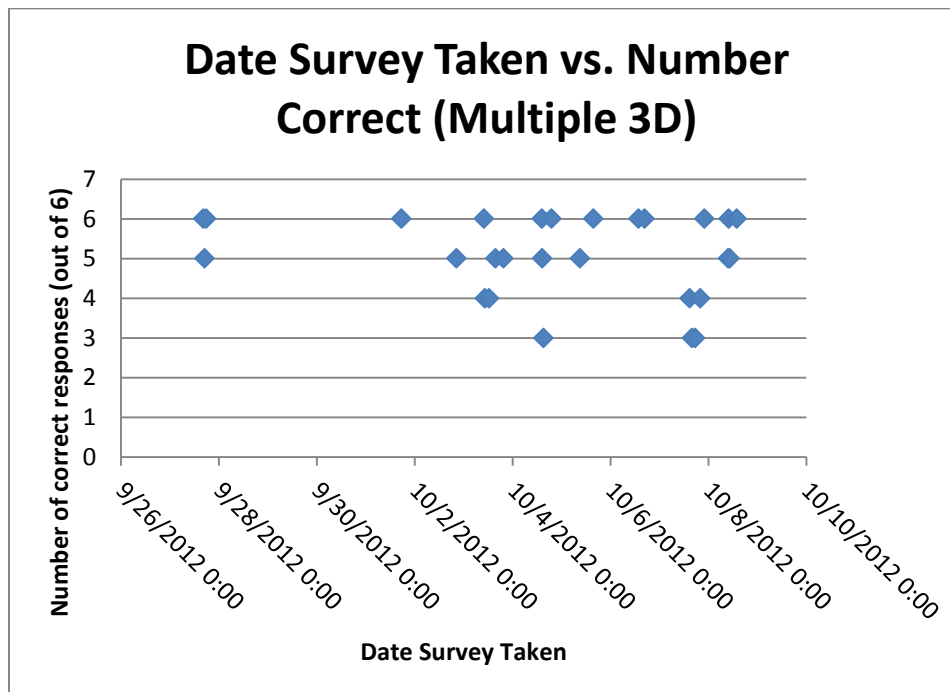
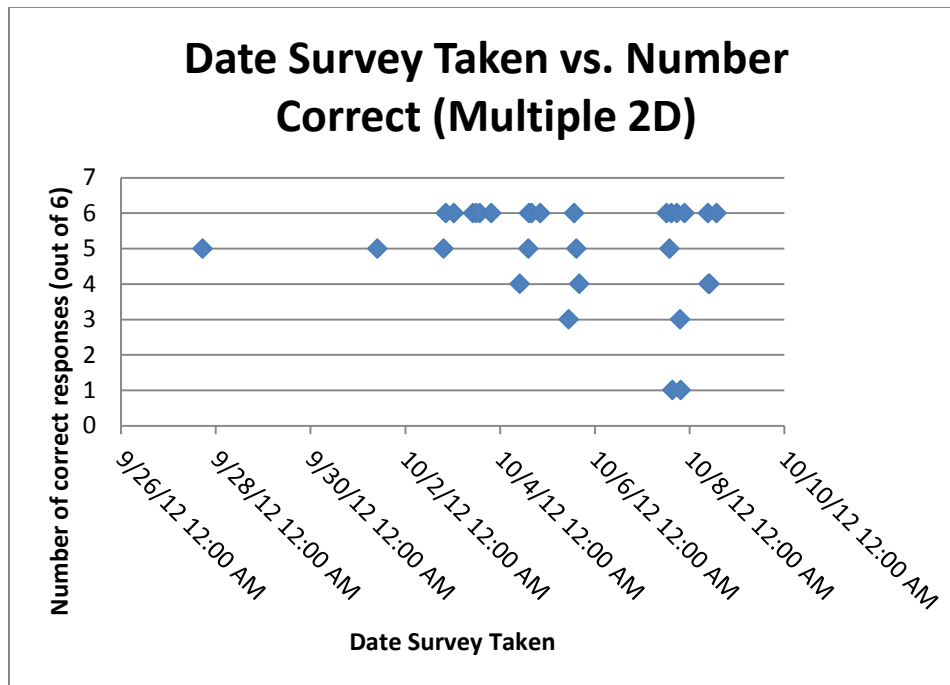


Figure E3: Response Time vs. Number Correct

Collection of timestamps for survey data allows comparison of the number of correct answers with the date the survey was taken. This comparison is shown in figure E4.



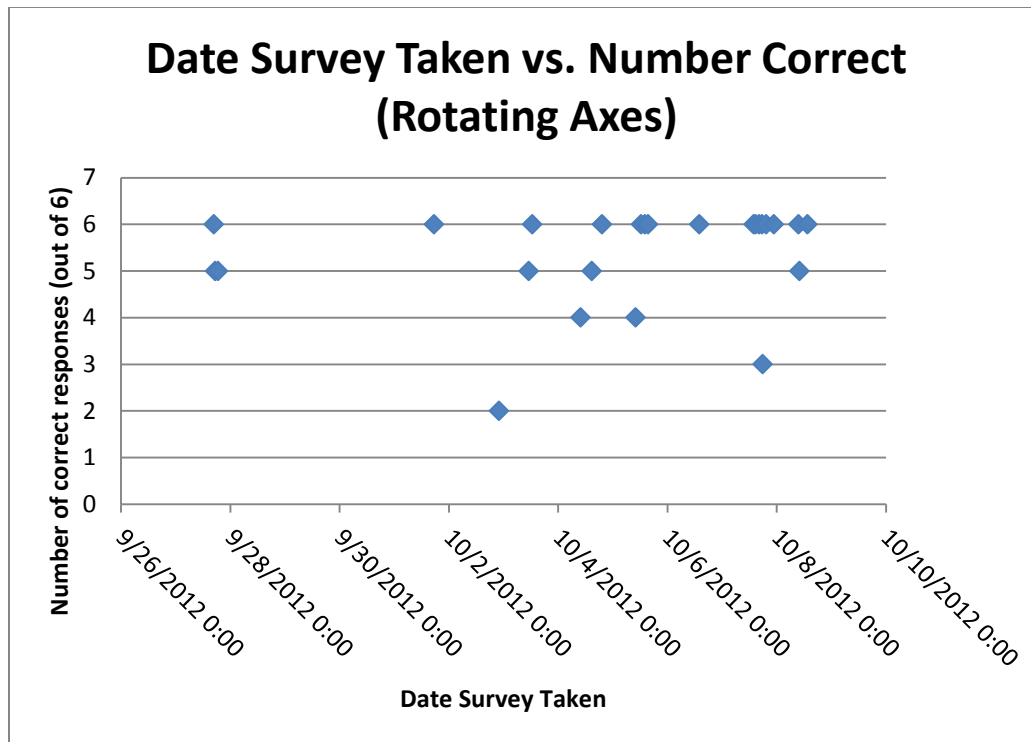
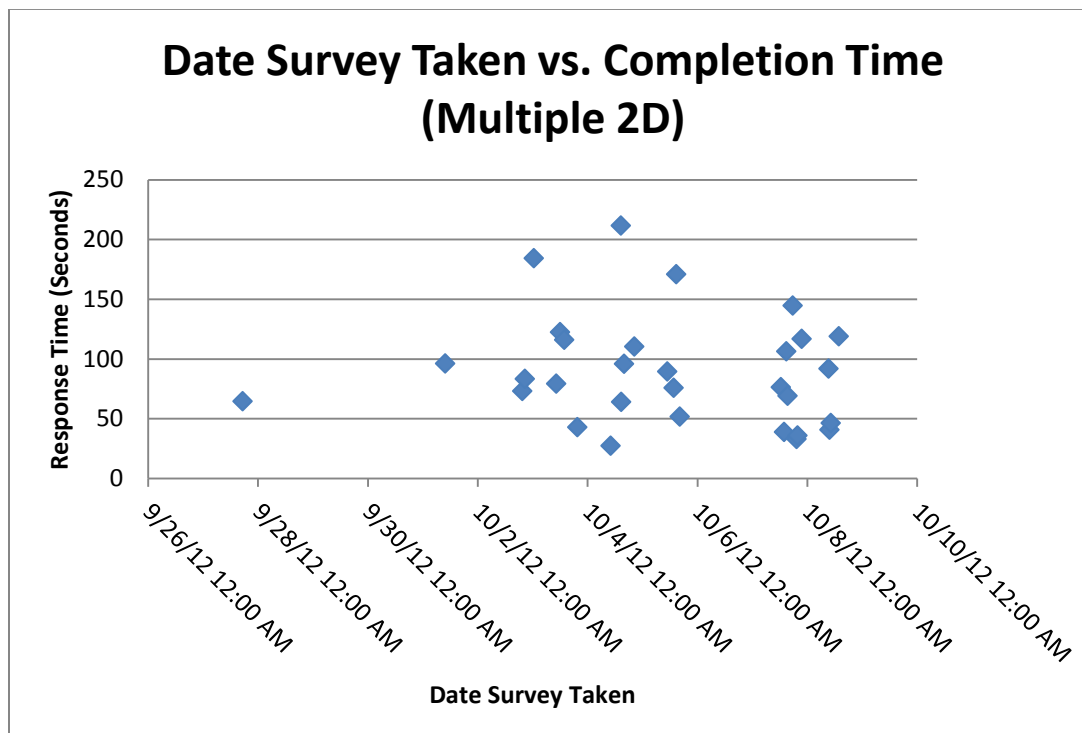


Figure E4: Date Survey Taken versus Number of Correct Answers

The date of survey completion compared to the time required to complete the survey is shown in figure E5.



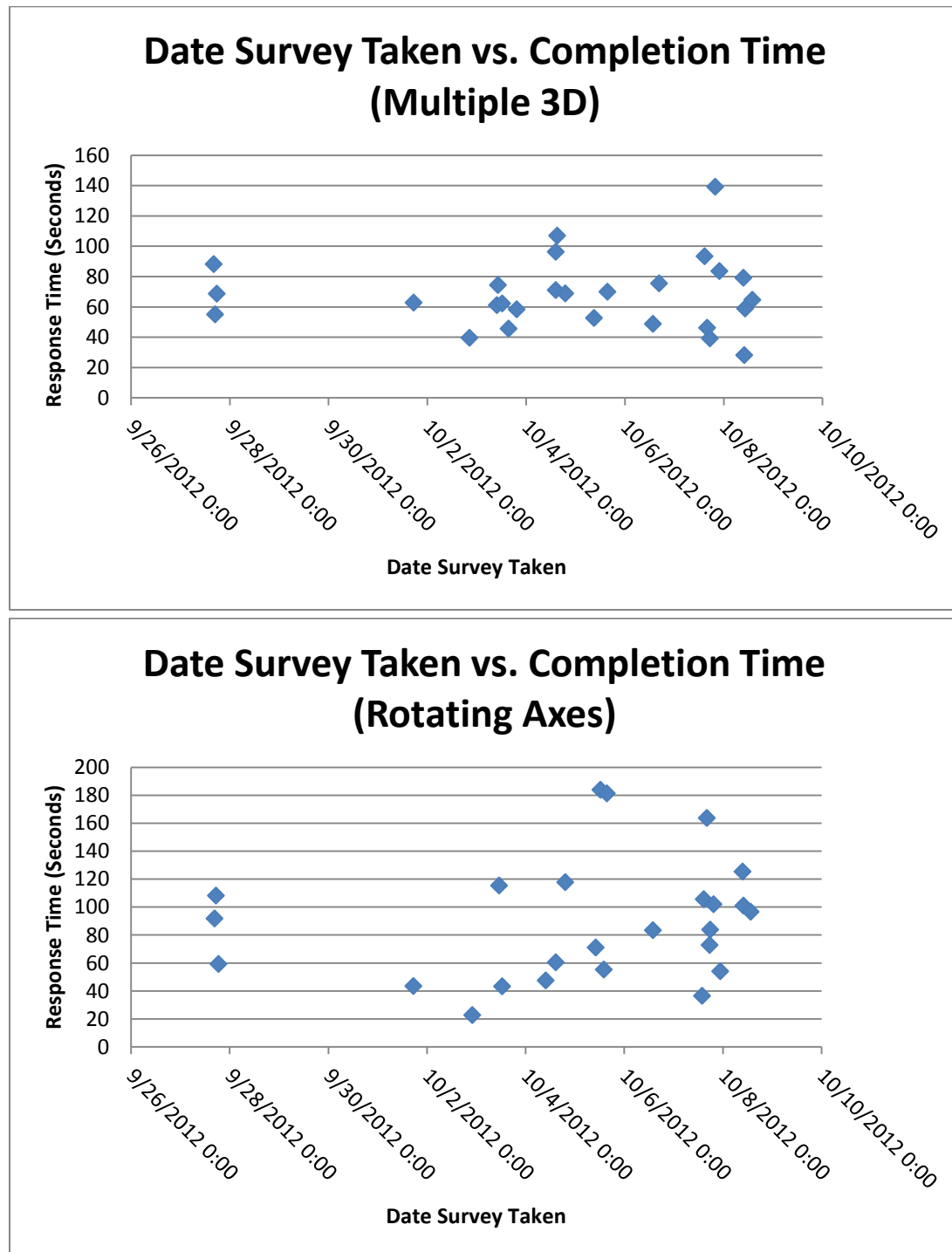


Figure E5: Date Survey Taken vs. Time required for Completion

The average time required for completion of the survey based on the number of correct answers given is shown in table E2.

| Measurement                         | Multiple 2D | Multiple 3D | Rotating Axes |
|-------------------------------------|-------------|-------------|---------------|
| Average time (0 correct) in seconds | No Data     | No Data     | No Data       |
| Average time (1 correct) in seconds | 52.3235     | No Data     | No Data       |
| Average time (2 correct) in seconds | No Data     | No Data     | 22.732        |
| Average time (3 correct) in seconds | 61.0235     | 64.14033    | 83.74         |
| Average time (4 correct) in seconds | 41.3865     | 92.3585     | 59.294        |
| Average time (5 correct) in seconds | 109.0665    | 57.556      | 88.8146       |
| Average time (6 correct) in seconds | 101.7121    | 68.07008    | 97.28219      |

Table E2: Time required for completion based on number of correct answers (fall data)

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